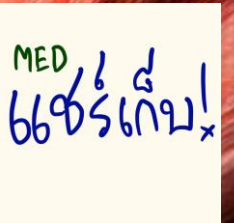


# BETA BLOCKER

———— Comprehensive review ————



# *Beta blocker*

- Block beta receptor
  - Sympathetic receptor blocker
- 1<sup>st</sup> Beta blocker: Pronethalol
  - By Sir James black, Nobel prize winner
  - Withdrawn from market due to potential association with tumor
- 1<sup>st</sup> Widely used Beta blocker: Propranolol
  - By Sir James black

# *Adrenergic receptor*

G protein coupled receptors  
targeted by catecholamines

- Subtypes
  - Alpha receptor ( $\alpha$ )
    - Alpha1: Mainly vessel
      - Vasoconstriction
    - Alpha2: Mainly nerve
      - Inhibit sympathetic output > Vasodilation
  - Beta receptor ( $\beta$ )
    - Beta1: Mainly heart
      - Positive inotropism and chronotropism
    - Beta2: Mainly Lung
      - Bronchodilatation
    - Beta3: Adipose tissue
      - Lipolysis and thermogenesis

**Table 1. Responses stimulated by  $\beta$ -adrenergic receptors in distinctive tissues.**

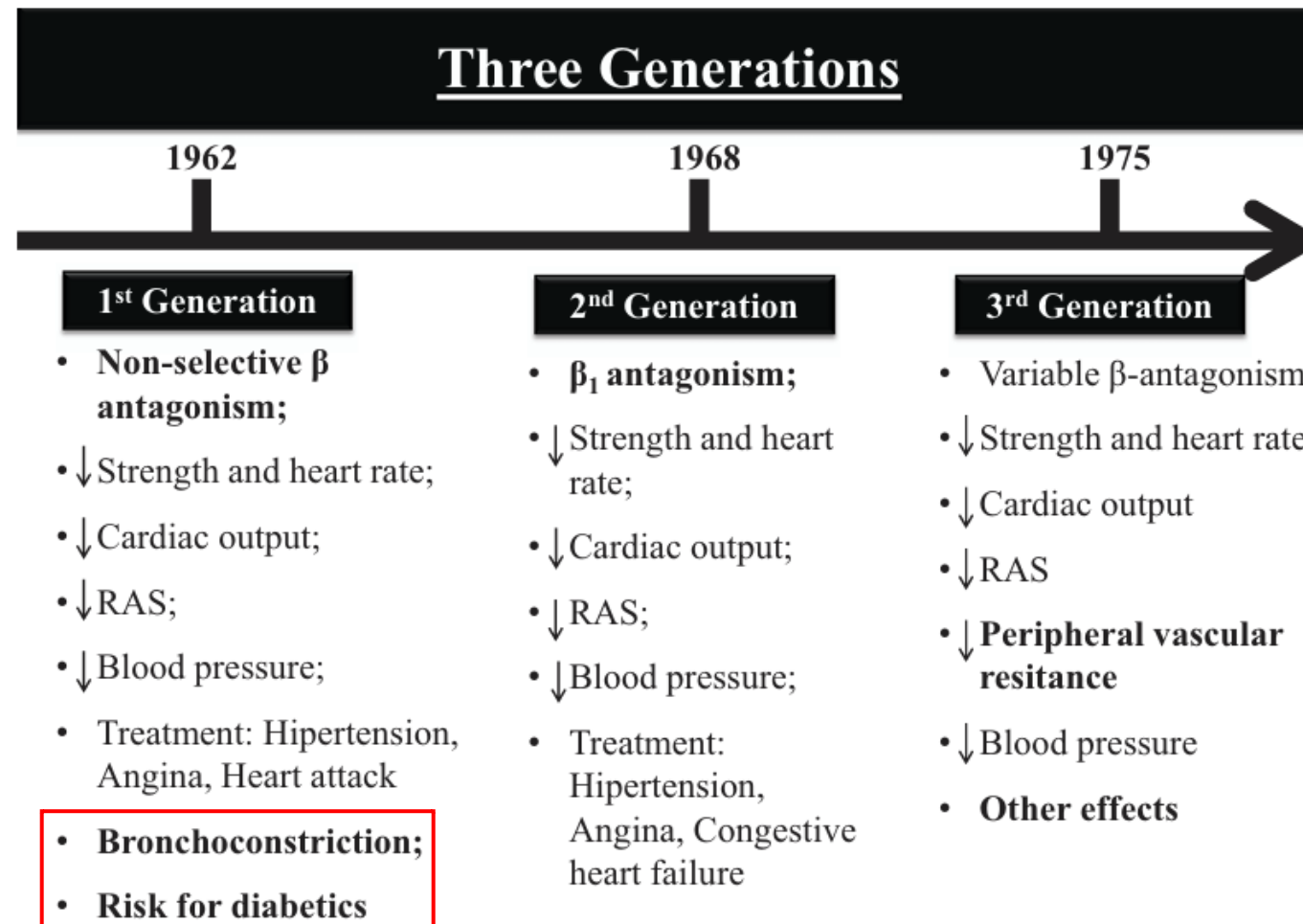
| Tissue/Receptor                      | $\beta_1$ [21-23]                     | $\beta_2$ [24-26]                     | $\beta_3$ [27]              |
|--------------------------------------|---------------------------------------|---------------------------------------|-----------------------------|
| <b>G Protein Activation</b>          | $G_s$                                 | $G_i$ and $G_s$                       | $G_i$                       |
| <b>Adipocytes</b>                    | -                                     | -                                     | Lipolysis and thermogenesis |
| <b>Heart</b>                         | Positive inotropism and chronotropism | Positive inotropism and chronotropism | Negative inotropism         |
| <b>Ileo and colon</b>                | -                                     | -                                     | Relaxation                  |
| <b>Lung</b>                          | -                                     | Relaxation                            | Relaxation                  |
| <b>Peripheral sympathetic nerves</b> | Release of norepinephrine             | Release of norepinephrine             | -                           |
| <b>Kidney justaglomerular cells</b>  | Secretion of renin                    | -                                     | -                           |
| <b>Vascular</b>                      | Relaxation                            | Relaxation                            | Relaxation                  |

| Receptor Type | Tissue Distribution                                                              | Mechanism of Action                                           | Agonist Potency         | Physiological Effects                                         | Agonist                                                                                 | Antagonist                                    |
|---------------|----------------------------------------------------------------------------------|---------------------------------------------------------------|-------------------------|---------------------------------------------------------------|-----------------------------------------------------------------------------------------|-----------------------------------------------|
| $\alpha_1$    | Vascular Smooth Muscles, Visceral smooth Muscles                                 | Gq-protein coupled activates Phospholipase C, IP3+DAG         | Epi $\geq$ NE $\gg$ Iso | Smooth muscle contractions, Gluconeogenesis, Vasoconstriction | Norepinephrine, Phenylephrine, Methoxamine                                              | Doxazosin, Phentolamine, Prazosin             |
| $\alpha_2$    | Pre-synaptic terminals, pancreas, platelets, Ciliary epithelium, Salivary Glands | Gi-protein coupled inhibits Adenyl cyclase                    | Epi $\geq$ NE $\gg$ Iso | Inhibits release of Neurotransmitter                          | Clonidine, Monoxidine                                                                   | Yohimbine, Idazoxan, Tolazoline               |
| $\beta_1$     | Heart, Kidney, some pre-synaptic terminals                                       | Gs-protein coupled activates Adenyl cyclase +PKA              | Iso $>$ Epi $\geq$ NE   | Increase heart rate and Renin secretion                       | Isoproterenol, Norepinephrine, Dobutamine                                               | Propranolol, Metoprolol, Atenolol             |
| $\beta_2$     | Visceral smooth muscles, Bronchioles, Liver, Skeletal Muscles                    | Gs-protein coupled activates Adenyl cyclase +PKA, Ca-channels | Iso $>$ Epi $\gg$ NE    | Vasodilation, Bronchodilation, Inhibits insulin secretion     | Isoproterenol, Salbutamol, Salmeterol, Albuterol, Formoterol, Terbutaline, Levalbuterol | Propranolol, ICI-118,551, Nadolol, Butoxamine |
| $\beta_3$     | Adipose Tissue                                                                   | Gs-protein coupled activates Adenyl cyclase +PKA              | Iso = NE $>$ Epi        | Increase lipolysis                                            | Isoproterenol, Amibegron, Solabegron                                                    | SR59230A                                      |

NE: Norepinephrine, Epi: Epinephrine and Iso: Isoproterenol

# *Generation of beta- blocker*

- **1<sup>st</sup> generation**
  - Non-selective beta blocker
    - Blocking both  $\beta_1$ - and  $\beta_2$  – Adrenoceptors
    - E.g. Propranolol
- **2<sup>nd</sup> generation**
  - Cardio-selective betablocker
    - Mainly block  $\beta_1$  adrenoceptor
    - E.g. Bisoprolol, Metoprolol, Atenolol, Esmolol
- **3<sup>rd</sup> generation**
  - Betablocker with additional effect
    - Alpha $\alpha_1$  blocker: Carvedilol, Labetalol
    - NO release: Nebivolol



**Fig. (2). Summary of the evolution of  $\beta$ -blockers and their basic differences.** The therapeutic effects and disadvantages of first-generation  $\beta$ -blockers relative to the second and third generations are shown in bold. For second-generation  $\beta$ -blockers, therapeutic effects and advantages over representatives of the first generation are shown in bold. Finally, the therapeutic effects and activity on  $\beta_1$ -adrenergic receptor are shown in bold for the third-generation  $\beta$ -blockers as an advantage over the two previous generations.

**Table II.** Pharmacologic Characteristics of  $\beta$ -Blockers<sup>3,7-9</sup>

| DRUG                                                                                                                                                | $\beta_1/\beta_2$ -SELECTIVITY | ISA | LIPOPHILICITY | MSA | ANCILLARY EFFECTS                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-----|---------------|-----|-----------------------------------------------------------------------------------|
| Nonselective $\beta$ -adrenergic antagonists                                                                                                        |                                |     |               |     |                                                                                   |
| Carteolol                                                                                                                                           | 0                              | +   | Low           | 0   | N/A                                                                               |
| Nadolol                                                                                                                                             | 0                              | 0   | Low           | 0   | N/A                                                                               |
| Penbutolol                                                                                                                                          | 0                              | +   | Moderate      | 0   | N/A                                                                               |
| Pindolol                                                                                                                                            | 0                              | ++  | High          | +   | N/A                                                                               |
| Propranolol                                                                                                                                         | 0                              | 0   | High          | ++  | N/A                                                                               |
| Sotalol                                                                                                                                             | 0                              | 0   | Low           | 0   | Additional antiarrhythmic effects                                                 |
| Timolol                                                                                                                                             | 0                              | 0   | High          | 0   |                                                                                   |
| Selective $\beta$ -adrenergic antagonists                                                                                                           |                                |     |               |     |                                                                                   |
| Acebutolol                                                                                                                                          | +                              | +   | Moderate      | +   | N/A                                                                               |
| Atenolol                                                                                                                                            | +                              | 0   | Low           | 0   | N/A                                                                               |
| Betaxolol                                                                                                                                           | ++                             | 0   | Moderate      | 0   | N/A                                                                               |
| Bisoprolol                                                                                                                                          | ++                             | 0   | Moderate      | 0   | N/A                                                                               |
| Esmolol (IV only)                                                                                                                                   | ++                             | 0   | Low           | 0   | N/A                                                                               |
| Metoprolol                                                                                                                                          | ++                             | 0   | High          | 0   | N/A                                                                               |
| $\alpha_1$ -Adrenergic and $\beta$ -adrenergic antagonists                                                                                          |                                |     |               |     |                                                                                   |
| Labetalol                                                                                                                                           | +                              | 0   | Low           | 0   | $\alpha_1$ -adrenergic-blocking activity; direct $\beta_2$ -vasodilatory activity |
| Carvedilol                                                                                                                                          | 0                              | 0   | Moderate      | ++  |                                                                                   |
| Selective $\beta$ -adrenergic antagonists with ancillary properties                                                                                 |                                |     |               |     |                                                                                   |
| Nebivolol                                                                                                                                           | +++                            | 0   | Moderate      | 0   | Endothelium-dependent, NO-mediated vasodilation                                   |
| Abbreviations: ISA, intrinsic sympathomimetic activity; IV, intravenous; MSA, membrane-stabilizing activity; N/A, not applicable; NO, nitric oxide. |                                |     |               |     |                                                                                   |

**Table 1** Cardioselectivity of  $\beta$ -blockers

| <b>Cardioselective<br/><math>\beta</math>-blockers</b> | <b>Relative <math>\beta_1/\beta_2</math><br/>selectivity</b> | <b>Usual dose<br/>range (mg/day)<br/>(frequency per day)</b> |
|--------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|
| Celiprolol                                             | 69                                                           | 200–400 (1)                                                  |
| Metoprolol                                             | 74                                                           | 50–200 (1–2)                                                 |
| Betaxolol                                              | 93                                                           | 10–20 (1)                                                    |
| Bisoprolol                                             | 103                                                          | 2.5–10 (1)                                                   |
| Nebivolol                                              | 321                                                          | 5–40 (1)                                                     |

# *Propranolol*

- 1st generation beta blocker
- 1<sup>st</sup>  $\beta$ -blocker widely used in the clinical practice
- High lipophilicity: Easily cross BBB
  - Pros: Good for migraine treatment
  - Cons: CNS side effect >> Depression, Insomnia
- Side effect
  - Beta1 blockage: Bronchospasm
  - Sympathetic blockage: New onset DM, DLP
- Current clinical indication
  - Migraine, Essential tremor, Hyperthyroidism, Portal hypertension

# *Metoprolol*

- 2nd generation beta blocker
- Cardio-selective to prevent beta2 associated side effects
- High lipophilicity: Easily cross BBB
  - Pros: Good for migraine treatment
  - Cons: CNS side effect >> Depression, Insomnia
- Two formula
  - Metoprolol tartrate (Immediate release): Rapid onset
  - Metoprolol succinate (Extended release): Longer effect
- Current main indication: Arrhythmia

# *Atenolol*

- 2nd generation beta blocker
- Cardio-selective to prevent beta2 associated side effects
- Low lipophilicity > Low CNS effect
- Current main indication:  
Arrhythmia

# *Bisoprolol*

- 2nd generation beta blocker
- Strong Cardio-selective beta-blocker
  - High  $\beta_1$  Selectivity
  - Third highest  $\beta_1$  Selectivity
- Low lipophilicity > Low CNS effect
- Current main indication:  
Arrhythmia, HFrEF

# *Esmolol*

- IV 2nd generation beta blocker
- Ultra-short half-life with rapid onset of action
  - Rapid onset 1-2 min: immediate control of conditions requiring quick beta-blockade
    - Very effective in Arrhythmias
  - Only in IV form
- Strong Cardio-selective beta-blocker
  - High  $\beta_1$  Selectivity
  - Second Highest  $\beta_1$  Selectivity

# *Labetalol*

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- 3rd generation betablocker
  - First representative of 3rd generation BB
- Non selective Betablocker with additional alpha-blocking effect
  - Vasodilatory beta blocker
- Rapid onset betablocker (5-10 min)
  - Available in PO, IV
- Very potent IV betablocker
  - Commonly used in HT Emergency, preeclampsia, Stroke
- Pregnancy category C: 1<sup>st</sup> line in preeclampsia

**Table 12. Hypertensive Emergencies Requiring Immediate BP Lowering**

| Clinical Presentation                                                                                | Timeline and Target BP                           | First Line Treatment                                       | Alternative                   |
|------------------------------------------------------------------------------------------------------|--------------------------------------------------|------------------------------------------------------------|-------------------------------|
| Malignant hypertension with or without TMA or acute renal failure                                    | Several hours, MAP –20% to –25%                  | Labetalol<br>Nicardipine                                   | Nitroprusside<br>Urapidil     |
| Hypertensive encephalopathy                                                                          | Immediate, MAP –20% to –25%                      | Labetalol<br>Nicardipine                                   | Nitroprusside                 |
| Acute ischaemic stroke and SBP >220 mm Hg or DBP >120 mm Hg                                          | 1 h, MAP –15%                                    | Labetalol<br>Nicardipine                                   | Nitroprusside                 |
| Acute ischaemic stroke with indication for thrombolytic therapy and SBP >185 mm Hg or DBP >110 mm Hg | 1 h, MAP –15%                                    | Labetalol<br>Nicardipine                                   | Nitroprusside                 |
| Acute hemorrhagic stroke and SBP >180 mm Hg                                                          | Immediate, 130<SBP<180 mm Hg                     | Labetalol<br>Nicardipine                                   | Urapidil                      |
| Acute coronary event                                                                                 | Immediate, SBP <140 mm Hg                        | Nitroglycerine<br>Labetalol                                | Urapidil                      |
| Acute cardiogenic pulmonary edema                                                                    | Immediate, SBP <140 mm Hg                        | Nitroprusside or nitroglycerine (with loop diuretic)       | Urapidil (with loop diuretic) |
| Acute aortic disease                                                                                 | Immediate, SBP <120 mm Hg and heart rate <60 bpm | Esmolol and nitroprusside or nitroglycerine or nicardipine | Labetalol or metoprolol       |
| Eclampsia and severe preeclampsia/HELLP                                                              | Immediate, SBP <160 mm Hg and DBP <105 mm Hg     | Labetalol or nicardipine and magnesium sulphate            |                               |

Adapted from van den Born et al.<sup>127</sup>

# *Labetalol*

- Labetalol IV solution is commonly available as 5 mg/mL
- Bolus dose
  - 20 mg IV over 2 min
  - May repeat with 40 mg IV
- Continuous dose (Conventional dose)
  - Preparation 1 mg/ml
    - Labetalol 100 mg (20 ml) + NSS up to 100 ml (80 ml)
  - Dosage: 1-8 mg/min
    - Labetalol 100 mg (20 ml) + NSS up to 100 ml (80 ml) IV drip 60 – 480 ml/hr
    - Titrate 1 mg/min (60 ml/hr) q 15 min

# *Carvedilol*

- 3<sup>rd</sup> generation betablocker
- Non selective Betablocker with additional alpha-blocking effect
  - Vasodilatory beta blocker
- Antioxidant effect
  - decreased plasma levels of 8-isoprostane and erythrocyte malondialdehyde
- Well known benefit for HFrEF
- Less metabolic profile complication
  - Less DM, DLP compared to older generation

# *Nebivolol*

- 3rd generation betablocker
- Non selective Betablocker with additional Nitric oxide (NO) releasing effect
  - Vasodilatory beta blocker
- Highest Beta1 selectivity beta blocker
- Less side effect
  - Increased NO effect
    - No erectile dysfunction (May improve ED)
  - Minimal metabolic profile complication
    - Less DM, DLP compared to older generation
  - Low CNS penetration
    - Minimal CNS side effect: Insomnia, depression

*Beta-blocker  
approved  
indication*

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# *Indication*

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- Hypertension
- Heart failure
- Coronary artery disease
- Arrhythmia
  - Atrial fibrillation
  - Supraventricular tachycardia
  - Ventricular tachycardia
- Migraine
- Essential tremor
- Portal hypertension



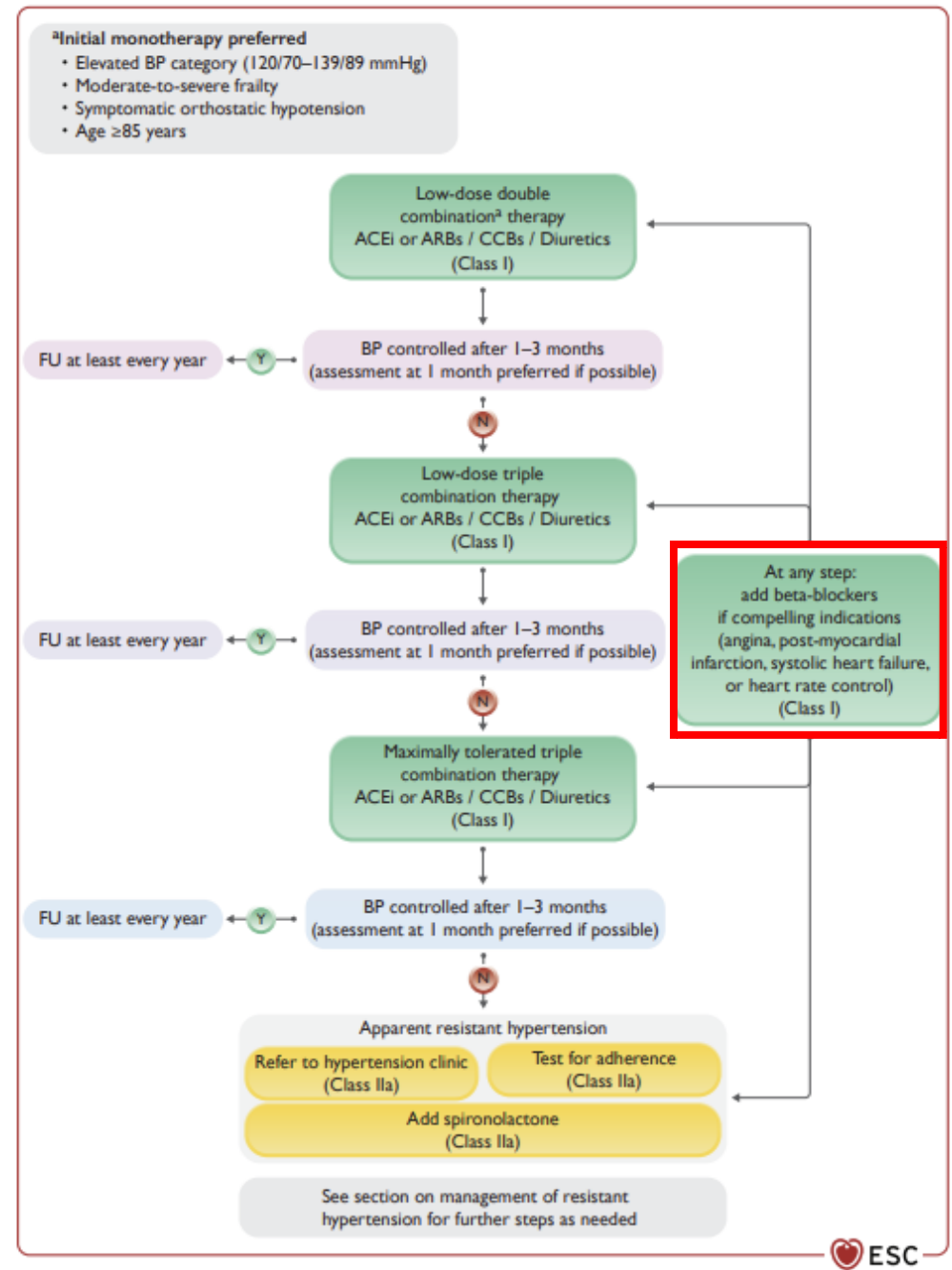
# *Hypertension*

# *Antihypertensive*

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- Historical main antihypertensive medication
  - 1<sup>st</sup> line in Thai hypertension guideline 2024
  - Replaced by newer generation in international guideline
- Highly recommended in patients with compelling indication
  - Angina, Post MI, Heart failure, HR control
- Preferably vasodilatory beta-blocker
  - High antihypertensive potency
  - Less metabolic side effect

# ESC 2024 Hypertension guideline



**Figure 18** Practical algorithm for pharmacological blood pressure lowering. ACEi, angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blocker; BP, blood pressure; CCB, calcium channel blocker; FU, follow-up.

# *Thai hypertension guideline 2024*

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- 1<sup>st</sup> line medication
  - ACEI/ARB
  - Beta blockers
  - Calcium channel blockers
  - Thiazide/Thiazide-like diuretics

# LIFE trial 2005

- P: Old HT (55-80 y) with LVH (by ECG)
- I: Losartan based antiHT
- C: Atenolol based antiHT
- O: Composite endpoint of cardiovascular morbidity and mortality

| Endpoint                     | Losartan (n=4605) |       | Atenolol (n=4588) |      | Adjusted hazard ratio (95% CI)† | p     | Unadjusted hazard ratio (95% CI) | p      |
|------------------------------|-------------------|-------|-------------------|------|---------------------------------|-------|----------------------------------|--------|
|                              | n                 | Rate* | n                 | Rate |                                 |       |                                  |        |
| Primary composite endpoint‡  | 508 (11%)         | 23.8  | 588 (13%)         | 27.9 | 0.87 (0.77–0.98)                | 0.021 | 0.85 (0.76–0.96)                 | 0.009  |
| Cardiovascular mortality     | 204 (4%)          | 9.2   | 234 (5%)          | 10.6 | 0.89 (0.73–1.07)                | 0.206 | 0.87 (0.72–1.05)                 | 0.136  |
| Stroke                       | 232 (5%)          | 10.8  | 309 (7%)          | 14.5 | 0.75 (0.63–0.89)                | 0.001 | 0.74 (0.63–0.88)                 | 0.0006 |
| Myocardial infarction        | 198 (4%)          | 9.2   | 188 (4%)          | 8.7  | 1.07 (0.88–1.31)                | 0.491 | 1.05 (0.86–1.28)                 | 0.628  |
| Other prespecified endpoints |                   |       |                   |      |                                 |       |                                  |        |
| Total mortality              | 383 (8%)          | 17.3  | 431 (9%)          | 19.6 | 0.90 (0.78–1.03)                | 0.128 | 0.88 (0.77–1.01)                 | 0.077  |
| Admitted to hospital for:    |                   |       |                   |      |                                 |       |                                  |        |
| Angina pectoris              | 160 (3%)          | 7.4   | 141 (3%)          | 6.6  | 1.16 (0.92–1.45)                | 0.212 | 1.13 (0.90–1.42)                 | 0.284  |
| Heart failure                | 153 (3%)          | 7.1   | 161 (4%)          | 7.5  | 0.97 (0.78–1.21)                | 0.765 | 0.95 (0.76–1.18)                 | 0.622  |
| Revascularisation            | 261 (6%)          | 12.2  | 284 (6%)          | 13.3 | 0.94 (0.79–1.11)                | 0.441 | 0.91 (0.77–1.08)                 | 0.292  |
| Resuscitated cardiac arrest  | 9 (0.2%)          | 0.4   | 5 (0.1%)          | 0.2  | 1.91 (0.64–5.72)                | 0.250 | 1.80 (0.60–5.36)                 | 0.294  |
| New-onset diabetes§          | 241 (6%)          | 13.0  | 319 (8%)          | 17.4 | 0.75 (0.63–0.88)                | 0.001 | 0.75 (0.63–0.88)                 | 0.001  |

\*Per 1000 patient-years of follow-up. †For degree of left ventricular hypertrophy and Framingham risk score at randomisation. ‡Cardiovascular mortality, stroke, and myocardial infarction (numbers of patients with a first primary event). §In patients without diabetes at randomisation (losartan, n=4019; atenolol, n=3979).

# ASCOT-BPLA trial 2005

- P: High risk HT (40-79 y) with risk factors
  - E.g. DM, Smoking, DLP
- I: Amlodipine based antiHT
- C: Atenolol based antiHT
- O: MI and fatal coronary heart disease (CHD)

|                                                              | Amlodipine-based regimen (n=9639) |               | Atenolol-based regimen (n=9618) |               | Unadjusted HR (95% CI) | p        |
|--------------------------------------------------------------|-----------------------------------|---------------|---------------------------------|---------------|------------------------|----------|
|                                                              | Number (%)                        | Rate per 1000 | Number (%)                      | Rate per 1000 |                        |          |
| <b>Primary endpoints</b>                                     |                                   |               |                                 |               |                        |          |
| Non-fatal myocardial infarction (including silent)+fatal CHD | 429 (5%)                          | 8.2           | 474 (5%)                        | 9.1           | 0.90 (0.79–1.02)       | 0.1052   |
| <b>Secondary endpoints</b>                                   |                                   |               |                                 |               |                        |          |
| Non-fatal myocardial infarction (excluding silent)+fatal CHD | 390 (4%)                          | 7.4           | 444 (5%)                        | 8.5           | 0.87 (0.76–1.00)       | 0.0458   |
| Total coronary endpoint                                      | 753 (8%)                          | 14.6          | 852 (9%)                        | 16.8          | 0.87 (0.79–0.96)       | 0.0070   |
| Total cardiovascular events and procedures                   | 1362 (14%)                        | 27.4          | 1602 (17%)                      | 32.8          | 0.84 (0.78–0.90)       | < 0.0001 |
| All-cause mortality                                          | 738 (8%)                          | 13.9          | 820 (9%)                        | 15.5          | 0.89 (0.81–0.99)       | 0.0247   |
| Cardiovascular mortality                                     | 263 (3%)                          | 4.9           | 342 (4%)                        | 6.5           | 0.76 (0.65–0.90)       | 0.0010   |
| Fatal and non-fatal stroke                                   | 327 (3%)                          | 6.2           | 422 (4%)                        | 8.1           | 0.77 (0.66–0.89)       | 0.0003   |
| Fatal and non-fatal heart failure                            | 134 (1%)                          | 2.5           | 159 (2%)                        | 3.0           | 0.84 (0.66–1.05)       | 0.1257   |

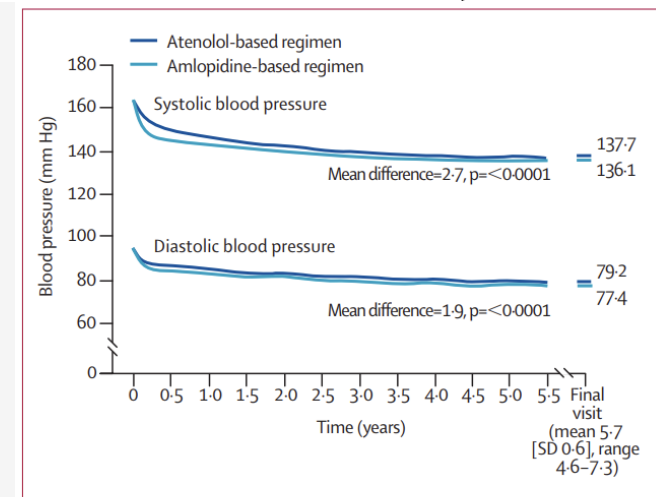


Figure 2: Blood pressure over time by group

*Dahlöf, B., et al. (2005). Prevention of cardiovascular events with an antihypertensive regimen of amlodipine adding perindopril as required versus atenolol adding bendroflumethiazide as required, in the Anglo-Scandinavian Cardiac Outcomes Trial-Blood Pressure Lowering Arm (ASCOT-BPLA): a multicentre randomised controlled trial. The Lancet, 366(9489), 895–906.*

# *Beta blocker antihypertensive dose*

| Beta-Blocker                    | Starting Dose       | Maximum Dose                 |
|---------------------------------|---------------------|------------------------------|
| Atenolol                        | 25–50 mg once daily | 100 mg once daily            |
| Bisoprolol                      | 2.5–5 mg once daily | 10 mg once daily             |
| Metoprolol Tartrate             | 50 mg twice daily   | 100–200 mg twice daily       |
| Metoprolol Succinate            | 25–50 mg once daily | 200 mg once daily            |
| Nebivolol                       | 5 mg once daily     | 40 mg once daily             |
| Propranolol (Immediate-Release) | 40 mg twice daily   | 160–320 mg per day (divided) |
| Propranolol (Extended-Release)  | 80 mg once daily    | 160–320 mg once daily        |
| Carvedilol                      | 6.25 mg twice daily | 25 mg twice daily            |
| Labetalol                       | 100 mg twice daily  | 400 mg twice daily           |



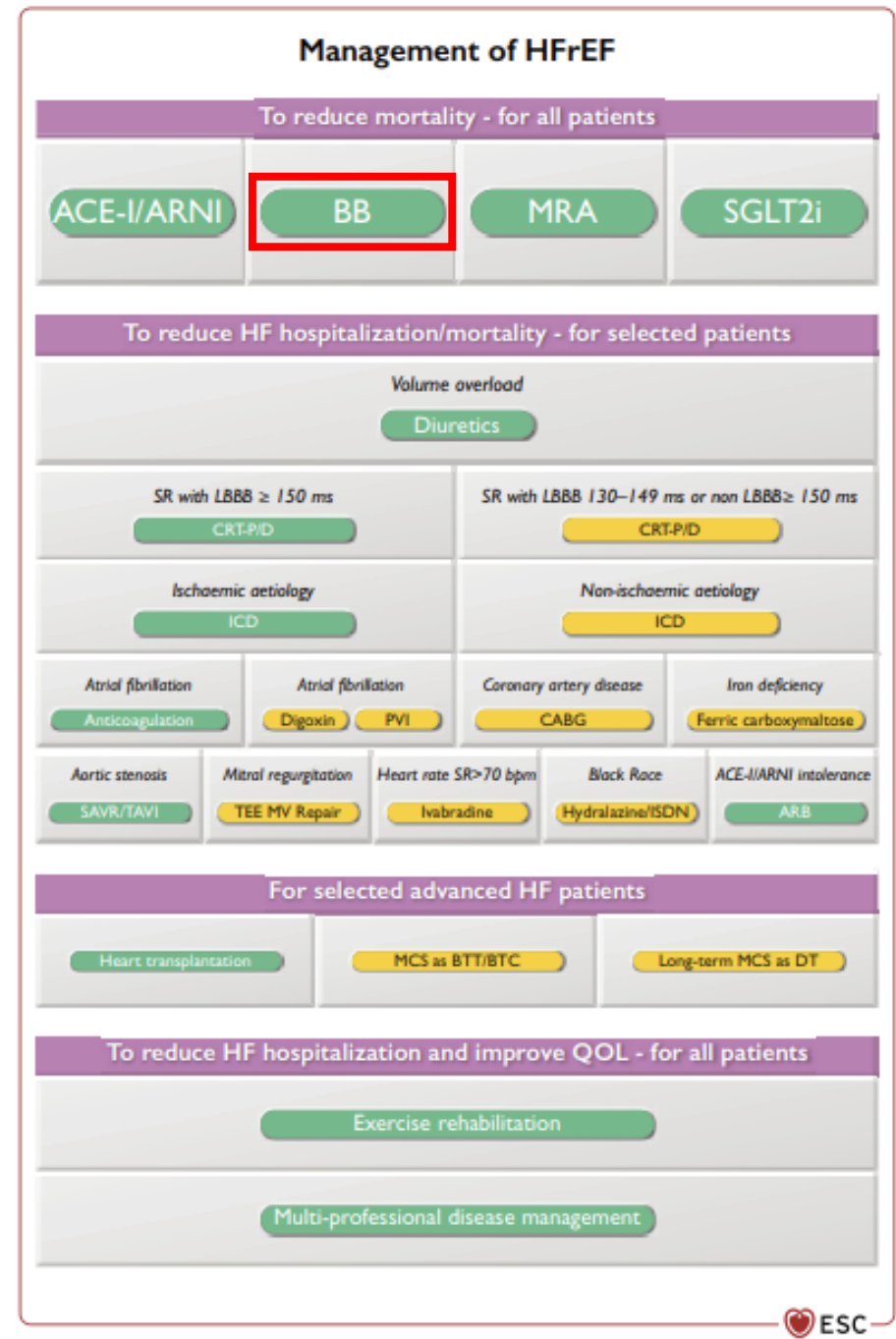
*Heart failure*

# *Heart failure*

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- Beta blockers is one of 4 pillars of HFrEF treatment
- Beta-blockers should be started at the lowest dose and titrated gradually to maximally tolerated dose
- Only 4 Beta-blockers have established RCT (Mnemonic: **C**ardiac **M**uscle **N**eurohormonal **B**locker)
  - **C**arvedilol
  - **M**etoprolol succinate
  - **N**ebivolol
  - **B**isoprolol

# ESC 2021 Heart failure guideline



# MERIT-HF trial 2000

- P: Symptomatic HFrEF (EF <40%)
  - on conventional therapy
    - ACEI, Diuretics
- I: Metoprolol CR/XL
  - 12.5 mg OD to 200 mg OD
- C: Placebo
- O: All-cause mortality

**Table 2.** Effect of Metoprolol CR/XL and Placebo on Combined End Points\*

| Combined End Points                                                                             | Metoprolol CR/XL Group, No. of Patients (n = 1990) | Placebo Group, No. of Patients (n = 2001) | Total | Risk Reduction, % (95% Confidence Interval) |
|-------------------------------------------------------------------------------------------------|----------------------------------------------------|-------------------------------------------|-------|---------------------------------------------|
| Total mortality or all-cause hospitalization                                                    | 641                                                | 767                                       | 1408  | 19 (10-27)                                  |
| Total mortality or hospitalization due to worsening heart failure                               | 311                                                | 439                                       | 750   | 31 (20-40)                                  |
| Death or heart transplantation                                                                  | 150                                                | 218                                       | 368   | 32 (16-45)                                  |
| Cardiac death or nonfatal acute myocardial infarction                                           | 139                                                | 225                                       | 364   | 39 (25-51)                                  |
| Total mortality or hospitalization or emergency department visit due to worsening heart failure | 318                                                | 455                                       | 773   | 32 (21-41)                                  |

\*Only the first end point that occurred in each patient was counted.  $P < .001$  for all comparisons. CR/XL indicates controlled release/extended release.

# CIBIS-II trial 1999

- P: Symptomatic HFrEF (EF <35%)
  - on conventional therapy
    - ACEI, Diuretics
- I: Bisoprolol
  - 1.25 mg OD to 10 mg OD
- C: Placebo
- O: All-cause mortality

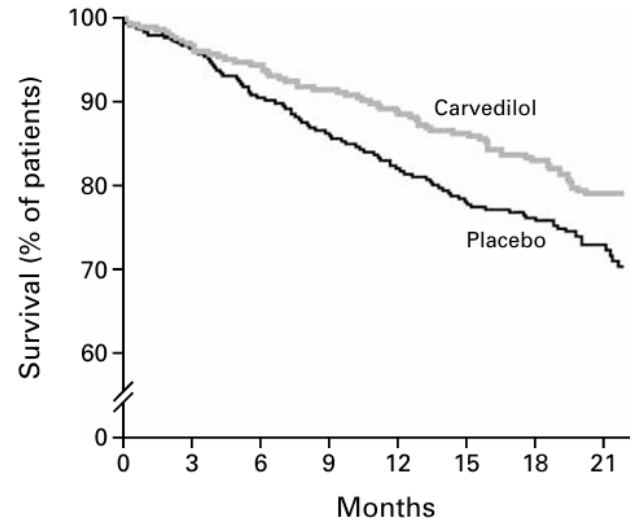
|                                                | Placebo<br>(n=1320) | Bisoprolol<br>(n=1327) | Hazard ratio<br>(95% CI) | p       |
|------------------------------------------------|---------------------|------------------------|--------------------------|---------|
| <b>Primary endpoint</b>                        |                     |                        |                          |         |
| All-cause mortality                            | 228 (17%)           | 156 (12%)              | 0.66 (0.54–0.81)         | <0.0001 |
| <b>Secondary endpoints</b>                     |                     |                        |                          |         |
| All-cause hospital admission                   | 513 (39%)           | 440 (33%)              | 0.80 (0.71–0.91)         | 0.0006  |
| All cardiovascular deaths                      | 161 (12%)           | 119 (9%)               | 0.71 (0.56–0.90)         | 0.0049  |
| Combined endpoint                              | 463 (35)            | 388 (29%)              | 0.79 (0.69–0.90)         | 0.0004  |
| Permanent treatment withdrawals                | 192 (15%)           | 194 (15%)              | 1.00 (0.82–1.22)         | 0.98    |
| <b>Exploratory analyses</b>                    |                     |                        |                          |         |
| Sudden death                                   | 83 (6%)             | 48 (4%)                | 0.56 (0.39–0.80)         | 0.0011  |
| Pump failure                                   | 47 (4%)             | 36 (3%)                | 0.74 (0.48–1.14)         | 0.17    |
| Myocardial infarction                          | 8 (1%)              | 7 (1%)                 | 0.85 (0.31–2.34)         | 0.75    |
| Other cardiovascular                           | 23 (2%)             | 28 (2%)                | 1.17 (0.67–2.03)         | 0.58    |
| Non-cardiovascular deaths                      | 18 (1%)             | 14 (1%)                | 0.75 (0.37–1.50)         | 0.41    |
| Unknown cause of death                         | 49 (4%)             | 23 (2%)                | 0.45 (0.27–0.74)         | 0.0012  |
| Hospital admission for worsening heart failure | 232 (18%)           | 159 (12%)              | 0.64 (0.53–0.79)         | 0.0001  |

Numbers refer to patients who presented at least once with given event. For hospital admissions, numbers refer to patients admitted at least once with any cause.

**Table 2: Primary and secondary endpoints and exploratory analyses**

# COPERNICUS Trial 2001

- P: Symptomatic HFrEF (EF <25%)
  - on conventional therapy
    - ACEI, Diuretics
- I: Carvedilol
  - Up to 25 mg BID
- C: Placebo
- O: All-cause mortality

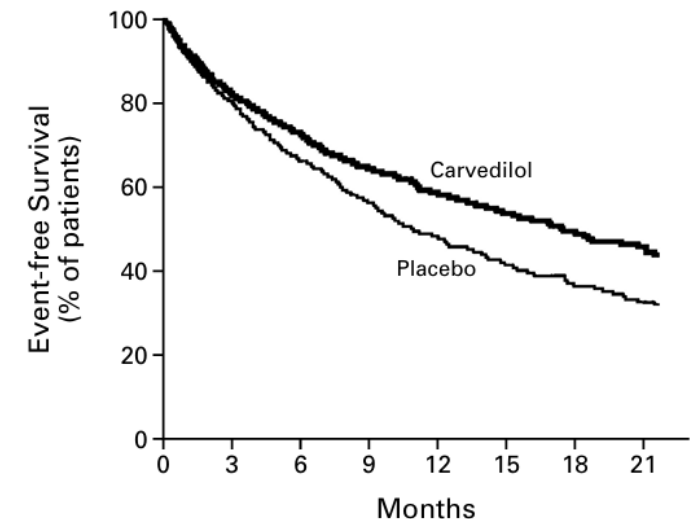


NO. OF PATIENTS AT RISK

|            |      |     |     |     |     |     |     |     |
|------------|------|-----|-----|-----|-----|-----|-----|-----|
| Placebo    | 1133 | 937 | 703 | 580 | 446 | 286 | 183 | 114 |
| Carvedilol | 1156 | 947 | 733 | 620 | 479 | 321 | 208 | 142 |

**Figure 1.** Kaplan-Meier Analysis of Time to Death in the Placebo Group and the Carvedilol Group.

The 35 percent lower risk in the carvedilol group was significant:  $P=0.00013$  (unadjusted) and  $P=0.0014$  (adjusted).



NO. OF PATIENTS AT RISK

|            |      |     |     |     |     |     |     |    |
|------------|------|-----|-----|-----|-----|-----|-----|----|
| Placebo    | 1133 | 767 | 513 | 377 | 262 | 154 | 88  | 55 |
| Carvedilol | 1156 | 789 | 559 | 431 | 318 | 208 | 122 | 81 |

**Figure 2.** Kaplan-Meier Analysis of Time to Death or First Hospitalization for Any Reason in the Placebo Group and the Carvedilol Group.

The 24 percent lower risk in the carvedilol group was significant ( $P<0.001$ ).

# COMET trial 2003

- P: Symptomatic HFrEF ( $\leq 35\%$ )
  - on conventional therapy
    - ACEI, Diuretics
- I: Carvedilol
- C: Metoprolol
- O: All-cause mortality

|                                            | Carvedilol (n=1511) | Metoprolol (n=1518) | Hazard ratio (95% CI) | p      |
|--------------------------------------------|---------------------|---------------------|-----------------------|--------|
| <b>All deaths</b>                          | 512 (34%)           | 600 (40%)           | 0.83 (0.74–0.93)      | 0.002  |
| 5 year Kaplan-Meier                        | 35.3%               | 41.0%               | ..                    | ..     |
| Time at risk (months)                      | 46.8 (18.9)         | 45.2 (19.5)         | ..                    | ..     |
| Yearly mortality rate                      | 8.3                 | 10.0                | ..                    | ..     |
| <b>Cardiovascular deaths</b>               | 438 (29%)           | 534 (35%)           | 0.80 (0.70–0.90)      | 0.0004 |
| <b>Non-cardiovascular deaths</b>           | 74 (5%)             | 66 (4%)             | 1.08 (0.77–1.50)      | ..     |
| <b>All deaths and all-cause admission*</b> | 1116 (74%)          | 1160 (76%)          | 0.94 (0.86–1.02)      | 0.122  |
| 5 year Kaplan-Meier                        | 75.5%               | 78.5%               | ..                    | ..     |
| Time at risk (months)                      | 27.1 (22.7)         | 25.9 (22.1)         | ..                    | ..     |
| First event admission                      | 1022 (68%)          | 1031 (68%)          | ..                    | ..     |
| First event death                          | 94 (6%)             | 129 (8%)            | ..                    | ..     |

Data are number of deaths (%) or mean (SD). \*Any patient counted once only.

Table 2: **Primary endpoints and components of the composite endpoint**

# SENIORS trial 2005

- P: Old Symptomatic CHF (Any EF, Age  $\geq 70$ )
  - on conventional therapy
    - ACEI, Diuretics
- I: Nebivolol
- C: Placebo
- O: All-cause mortality or CV hospitalization

**Table 3** Primary and main secondary outcomes (time to first event)<sup>a</sup>

|                                                                  | Nebivolol <sup>b</sup> (n = 1067) | Placebo <sup>b</sup> (n = 1061) | HR <sup>a</sup> | 95% CI    | P-value <sup>a</sup> |
|------------------------------------------------------------------|-----------------------------------|---------------------------------|-----------------|-----------|----------------------|
| <b>Primary outcome</b>                                           |                                   |                                 |                 |           |                      |
| All cause mortality or CV hospitalization                        | 332 (31.1%)/20.3                  | 375 (35.3%)/23.9                | 0.86            | 0.74–0.99 | 0.039                |
| All cause mortality contributing to primary outcome <sup>c</sup> | 76 (7.1%)/4.1                     | 99 (9.3%)/5.4                   |                 |           |                      |
| CV hospitalizations contributing to primary outcome <sup>c</sup> | 256 (24.0%)/16.3                  | 276 (26.0%)/18.3                |                 |           |                      |
| Primary outcome, unadjusted analysis                             | 332 (31.1%)/20.3                  | 375 (35.3%)/23.9                | 0.85            | 0.74–0.99 | 0.034                |
| <b>Secondary outcomes</b>                                        |                                   |                                 |                 |           |                      |
| All cause mortality                                              | 169 (15.8%)/9.1                   | 192 (18.1%)/10.4                | 0.88            | 0.71–1.08 | 0.21                 |
| CV mortality                                                     | 123 (11.5%)/6.9                   | 145 (13.7%)/8.2                 | 0.84            | 0.66–1.07 | 0.17                 |
| Sudden cardiac death <sup>c,d</sup>                              | 44 (4.1%)/2.5                     | 70 (6.6%)/4.0                   |                 |           |                      |
| Non-CV mortality <sup>c</sup>                                    | 26 (2.4%)/1.5                     | 20 (1.9%)/1.1                   |                 |           |                      |
| Unknown/not classified <sup>c,e</sup>                            | 20 (1.9%)/1.1                     | 27 (2.5%)/1.5                   |                 |           |                      |
| CV hospitalization                                               | 256 (24.0%)/16.3                  | 276 (26.0%)/18.3                | 0.90            | 0.76–1.06 | 0.20                 |
| CV mortality or CV hospitalization                               | 305 (28.6%)/19.4                  | 350 (33.0%)/23.2                | 0.84            | 0.72–0.98 | 0.027                |
| All cause hospitalization                                        | 359 (33.6%)/24.4                  | 364 (34.3%)/25.7                | 0.96            | 0.82–1.10 | 0.47                 |
| All cause mortality or all cause hospitalization                 | 408 (38.2%)/26.6                  | 443 (41.8%)/30.1                | 0.90            | 0.78–1.02 | 0.082                |

CV, cardiovascular.

<sup>a</sup>HR (and P-value) calculated on time to event. Analyses adjusted by gender, age, and left ventricular ejection fraction, unless otherwise specified.

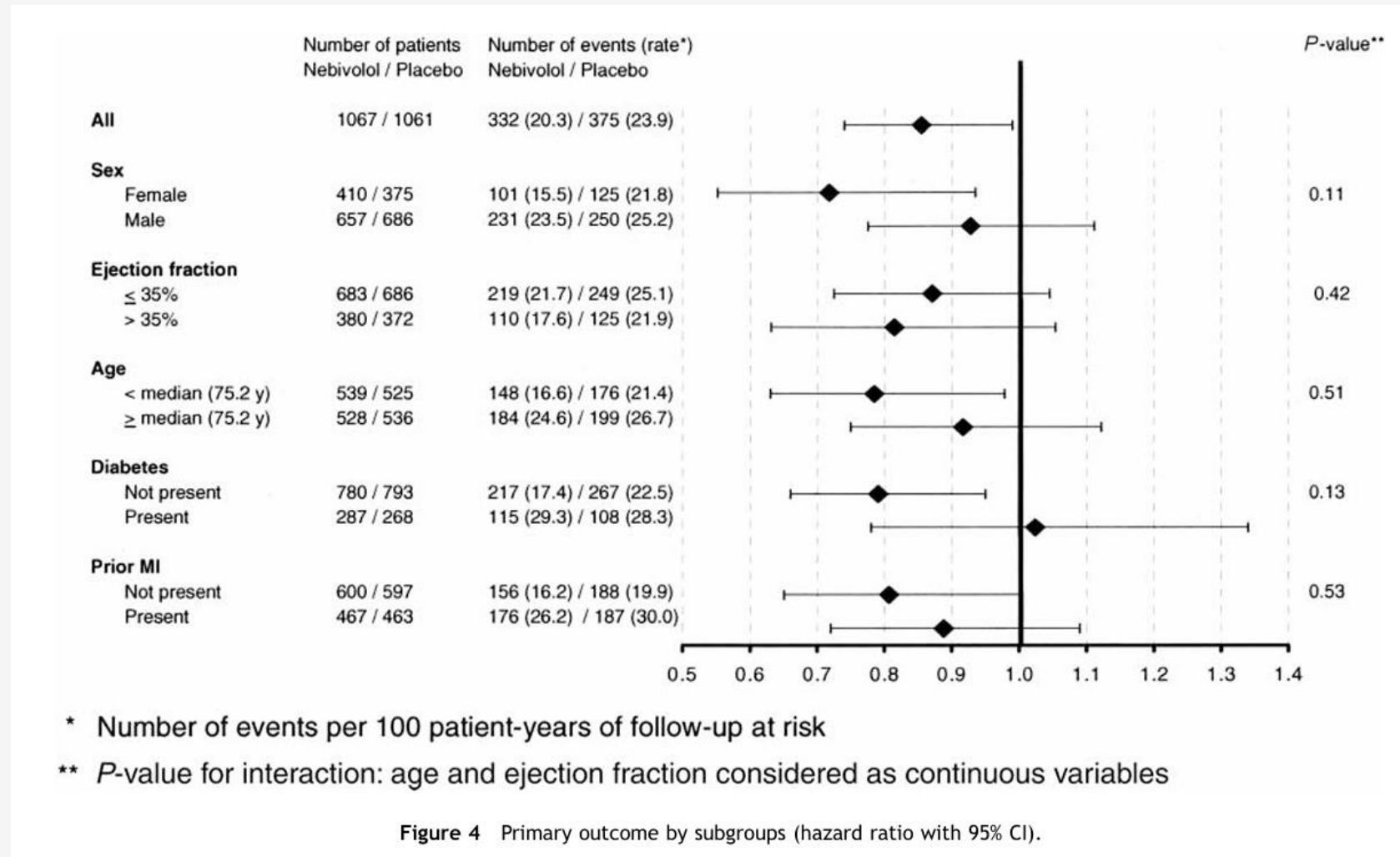
<sup>b</sup>Each cell in these columns contains the number of events, the percentage of events, and the annual rate as number of events per 100 patient-years of follow-up at risk.

<sup>c</sup>Not pre-specified analyses.

<sup>d</sup>Considered within CV mortality.

<sup>e</sup>Unknown/not classified category includes: not witnessed, not sudden cardiac death, not classified, vital status information from patients who prematurely terminated the study.

# SENIORS trial: Subgroup analysis



**TABLE 1** Characteristics of Major  $\beta$ -Blocker Trials in HF

|                                          | Study               |                         |                         |                           |                     |                        |
|------------------------------------------|---------------------|-------------------------|-------------------------|---------------------------|---------------------|------------------------|
|                                          | USCS<br>(N = 1,094) | MERIT-HF<br>(N = 3,991) | CIBIS-II<br>(N = 2,647) | COPERNICUS<br>(N = 2,289) | BEST<br>(N = 2,708) | SENIORS<br>(N = 2,128) |
| $\beta$ -blocker                         | Carvedilol          | Metoprolol Succinate    | Bisoprolol              | Carvedilol                | Bucindolol          | Nebivolol              |
| Mean age, yrs                            | 58                  | 64                      | 61                      | 64                        | 60                  | 76                     |
| Starting dose, mg                        | 6.25 b.i.d.         | 12.5 q.d.               | 1.25 q.d.               | 3.125 b.i.d.              | 3.0 b.i.d.          | 1.25 q.d.              |
| Target dose, mg                          | 25-50 b.i.d.        | 200 q.d.                | 10 q.d.                 | 25 b.i.d.                 | 50-100 b.i.d.       | 10 q.d.                |
| Mean daily dose achieved, mg             | 45.0                | 159.0                   | 7.5                     | 37.0                      | 152.0               | 7.7                    |
| Baseline heart rate, beats/min*          | 84 $\pm$ 12         | 83 $\pm$ 10             | 80 $\pm$ 15             | 83 $\pm$ 13               | 82 $\pm$ 13         | 79 $\pm$ 14            |
| Heart rate reduction, beats/min          | 12.6                | -14.0                   | -9.8                    | NR                        | -9.4                | -10.3                  |
| Baseline SBP, mm Hg*                     | 116 $\pm$ 17        | 130 $\pm$ 17            | 129 $\pm$ 19            | 123 $\pm$ 19              | 117 $\pm$ 18        | 139 $\pm$ 20           |
| Titration period, weeks                  | 2-10                | 1-8                     | 1-15                    | 1-8                       | 1-9                 | 1-16                   |
| % Relative effect on all-cause mortality | $\downarrow$ 65     | $\downarrow$ 34         | $\downarrow$ 34         | $\downarrow$ 35           | $\downarrow$ 10     | $\downarrow$ 12        |
| p value                                  | <0.001              | <0.001                  | <0.001                  | <0.001                    | 0.13                | 0.21                   |

\*Values are mean  $\pm$  SD.

$\downarrow$  = reduced; BEST = Beta-Blocker Evaluation of Survival Trial; b.i.d. = twice a day; CIBIS-II = Cardiac Insufficiency Bisoprolol Study II; COPERNICUS = Carvedilol Prospective Randomized Cumulative Survival; HF = heart failure; MERIT-HF = Metoprolol CR/XL Randomized Intervention Trial in Congestive Heart Failure; q.d. = daily; SBP = systolic blood pressure; SENIORS = Study of the Effects of Nebivolol Intervention on Outcomes and Rehospitalisation in Seniors with Heart Failure; USCS = U.S. Carvedilol HF Study.



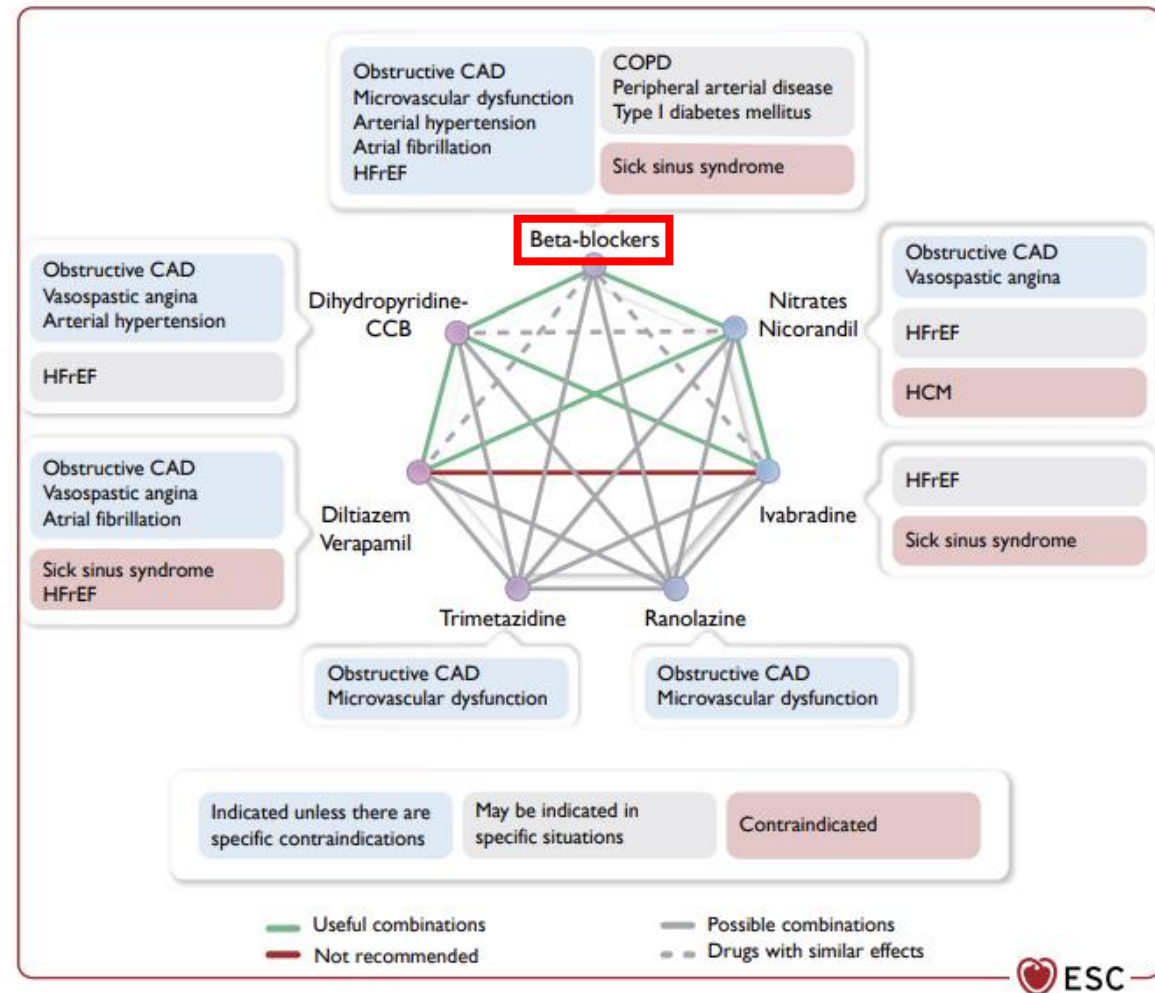
# *Coronary artery disease*

# *Coronary artery disease*

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- Benefits
  - Alleviate anginal chest pain
  - May reduce mortality post myocardial infarction
    - Controversial in post reperfusion era
- Effective in obstructive CAD
  - Less effective in vasospastic angina
- Heart rate goal: 55-60 bpm (ESC 2024)
- Highly recommended in presence of others compelling indication
  - HT, AF, HFrEF

# ESC CCS 2024 guideline



**Figure 9** Possible combinations of antianginal drugs. CAD, coronary artery disease; CCB, calcium channel blocker; COPD, chronic obstructive pulmonary disease; HCM, hypertrophic cardiomyopathy; HFrEF, heart failure with reduced ejection fraction. The schematic shows useful combinations (green lines), combinations that are not recommended (red lines), possible combinations (solid blue lines), and drugs with similar effects (blue dashed lines), which can be combined in selected indications: HFrEF (ivabradine and beta-blocker), atrial fibrillation (diltiazem/verapamil and beta-blocker), vasospastic angina (dihydropyridine CCB and nitrates). Modified from Davies et al.<sup>555</sup>.

# ACC/AHA CCS 2023 guideline

Virani, S. S., et al. (2023). 2023 AHA/ACC/ACCP/ASPC/NLA/PCNA Guideline for the Management of Patients with Chronic Coronary Disease: A report of the American Heart Association/American College of Cardiology Joint Committee on Clinical Practice Guidelines. *Circulation*, 148(9).

## 4.3.2. Beta Blockers

### Recommendations for Beta Blockers

Referenced studies that support the recommendations are summarized in the [Online Data Supplement](#).

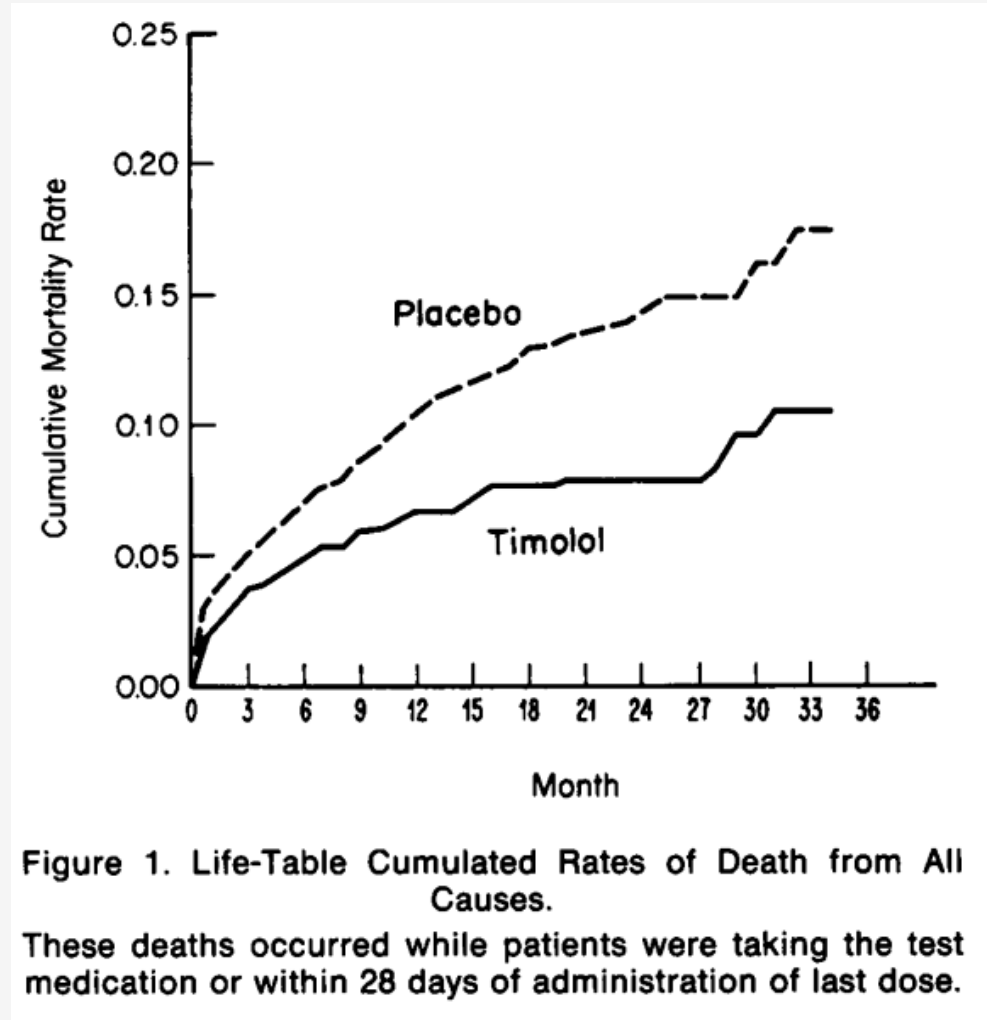
| COR           | LOE  | Recommendations                                                                                                                                                                                                                                                                                                                    |
|---------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1             | A    | 1. In patients with CCD and LVEF $\leq 40\%$ with or without previous MI, the use of beta-blocker therapy is recommended to reduce the risk of future MACE, including cardiovascular death. <sup>1-3</sup>                                                                                                                         |
| 1             | A    | 2. In patients with CCD and LVEF $< 50\%$ , the use of sustained release metoprolol succinate, carvedilol, or bisoprolol with titration to target doses is recommended in preference to other beta blockers. <sup>*1,3-8</sup>                                                                                                     |
| 2b            | B-NR | 3. In patients with CCD who were initiated on beta-blocker therapy for previous MI without a history of or current LVEF $\leq 50\%$ , angina, arrhythmias, or uncontrolled hypertension, it may be reasonable to reassess the indication for long-term ( $>1$ year) use of beta-blocker therapy for reducing MACE. <sup>9-15</sup> |
| 3: No Benefit | B-NR | 4. In patients with CCD without previous MI or LVEF $\leq 50\%$ , the use of beta-blocker therapy is not beneficial in reducing MACE, in the absence of another primary indication for beta-blocker therapy. <sup>†16-19</sup>                                                                                                     |

\*Modified from the 2022 AHA/ACC/HFSA Guideline for the Management of Heart Failure.<sup>20</sup>

†Adapted from the 2021 ACC/AHA/SCAI Guideline for Coronary Artery Revascularization.<sup>21</sup>

# *Timolol after myocardial infarction trial 1981*

- P: Post MI patients
- I: Timolol
- C: Placebo
- O: All-cause mortality



# BHAT trial 1982

- P: Post MI patients
- I: Propranolol
- C: Placebo
- O: All-cause mortality

(F/U 2-4 years, Average 25 mo)

Table 3.—Cause-Specific Mortality by Treatment Group

| Cause of Death                 | Propranolol   |              | Placebo       |              | P*<br>(Two-Sided) |
|--------------------------------|---------------|--------------|---------------|--------------|-------------------|
|                                | No. of Deaths | Mortality, % | No. of Deaths | Mortality, % |                   |
| Total mortality                | 138           | 7.2          | 188           | 9.8          | <.005             |
| Cardiovascular disease         | 127           | 6.6          | 171           | 8.9          | <.01              |
| Arteriosclerotic heart disease | 119           | 6.2          | 164           | 8.5          | <.01              |
| Sudden†                        | 64‡           | 3.3          | 89            | 4.6          | <.05              |
| Nonsudden                      | 55            | 2.9          | 75            | 3.9          | NS                |
| Other cardiovascular disease   | 8             | 0.4          | 7             | 0.4          | NS                |
| Noncardiovascular disease      | 11            | 0.6          | 17            | 0.9          | NS                |

\*Because of the numerous statistical tests performed, the *P* values for cause-specific mortality should be interpreted cautiously.

†Deaths occurring less than one hour from onset of symptoms.

‡One death coded as sudden occurred approximately one week after cardiac surgery.

## *Clinical Outcomes with $\beta$ -blockers for Myocardial Infarction A Meta-Analysis of Randomized Trials: Sripal Bangalore et al.*

- P: Post MI patients
  - Pre-reperfusion era trial
  - Post reperfusion era trials
- I: Beta-blocker
- C: No beta blocker
- O: All-cause mortality

| Outcomes          | Pre-Reperfusion Era (RR, 95% CI) | Reperfusion Era (RR, 95% CI) |
|-------------------|----------------------------------|------------------------------|
| Cardiac Death     | 0.87 (0.78, 0.98)                | 1.00 (0.91, 1.09)            |
| Sudden Death      | 0.77 (0.56, 1.05)                | 0.94 (0.86, 1.01)            |
| MI                | 0.78 (0.62, 0.97)                | 0.72 (0.62, 0.83)            |
| Angina Pectoris   | 0.88 (0.82, 0.95)                | 0.80 (0.65, 0.98)            |
| Stroke            | 2.96 (0.47, 18.81)               | 1.09 (0.91, 1.30)            |
| HF                | 1.06 (0.98, 1.16)                | 1.10 (1.05, 1.16)            |
| Cardiogenic Shock | 1.05 (0.89, 1.23)                | 1.29 (1.18, 1.40)            |
| Drug Withdrawal   | 1.13 (1.02, 1.24)                | 1.64 (1.55, 1.73)            |

# *Controversial issue*

- Beta blocker for post-MI
  - Historical evidences from pre-reperfusion era
    - Beta-blocker reduce mortality post MI
  - Post Reperfusion era
    - More standard medication for MI
      - E.g. Antithrombotic, Statin
    - Proven benefit in HFrEF (HFrEF trials)
    - Controversial benefit in non-HFrEF
      - Negative mortality benefit in some observational studies
- Risk of long-term beta-blocker usage
  - Fatigue, Depression, Drug-drug interaction



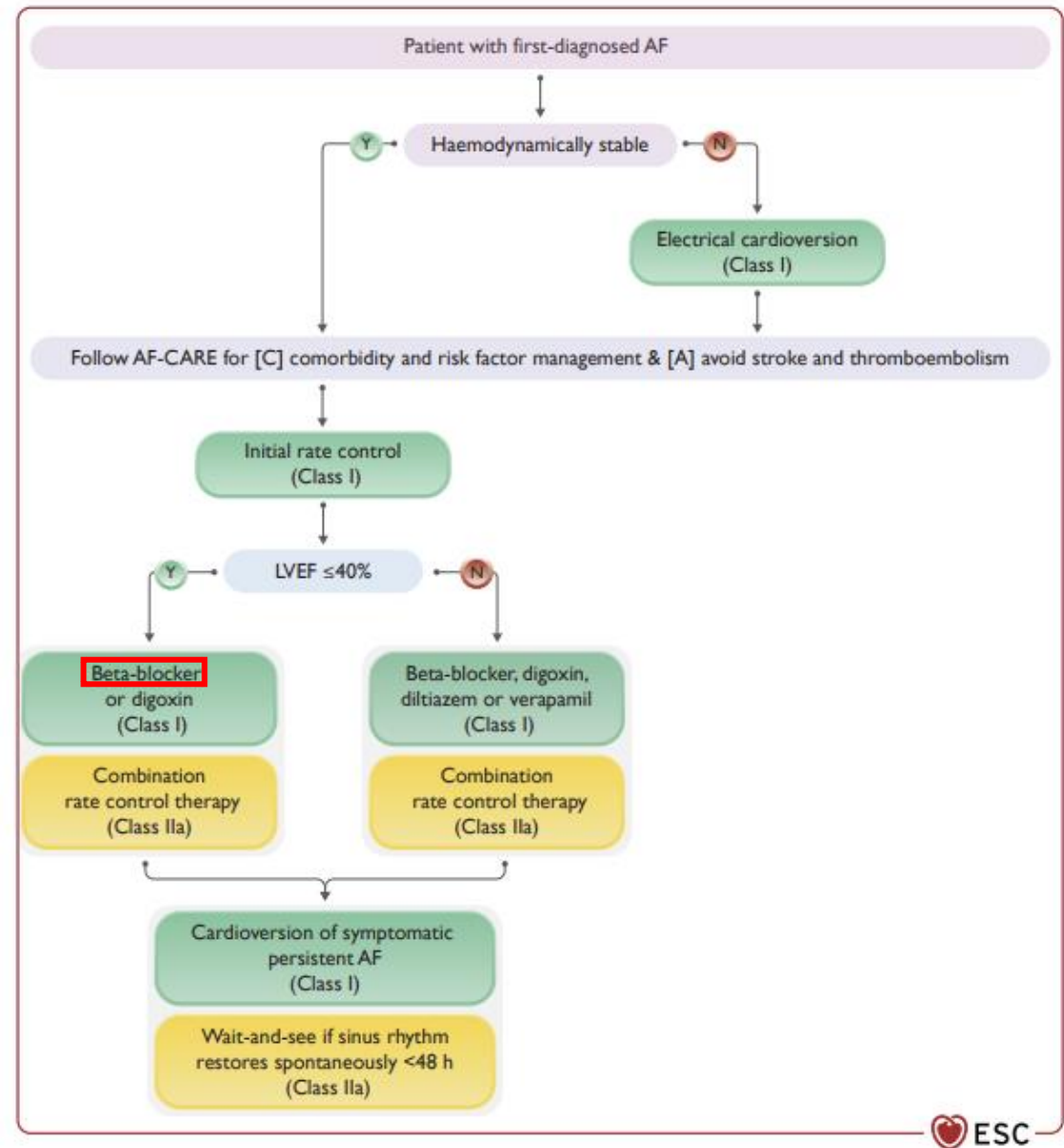
# *Atrial fibrillation*

# *Atrial fibrillation*

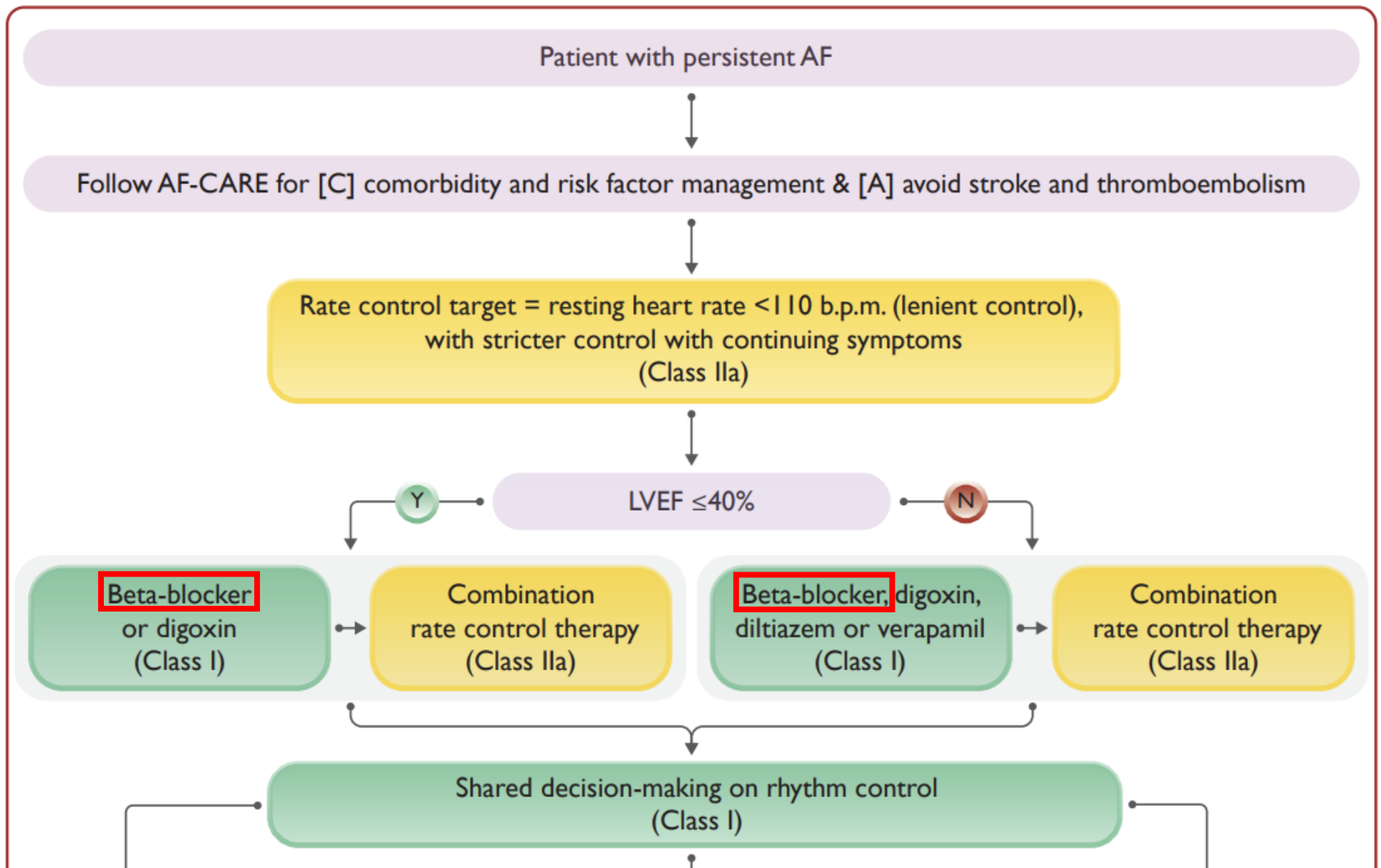
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- First line medication for AF rate control
- Effective in preventing recurrence of AF
- Initial HR goal: 110 bpm
- Rate control
  - Acute rate control
    - IV Beta blocker: Esmolol
      - Ultrarapid beta blocker
  - Chronic rate control
    - Oral beta blocker: Metoprolol, Atenolol, Bisoprolol, Carvedilol

# ESC AF 2024 guideline



**Figure 4** [R] Pathway for patients with first-diagnosed AF. AF, atrial fibrillation; AF-CARE, Atrial fibrillation—[C] Comorbidity and risk factor management, [A] Avoid stroke and thromboembolism, [R] Reduce symptoms by rate and rhythm control, [E] Evaluation and dynamic reassessment; LVEF, left ventricular ejection fraction. After following the pathway for first-diagnosed AF, patients with recurrent AF should enter the AF-CARE [R] pathway for paroxysmal, persistent, or permanent AF, depending on the type of their AF.



**Table 21. Pharmacological Agents for Rate Control in Patients With AF**

|                      | Intravenous Administration                                         | Oral Maintenance Dose                              | Elimination Half-Life                  | Notes              |
|----------------------|--------------------------------------------------------------------|----------------------------------------------------|----------------------------------------|--------------------|
| <b>Beta blockers</b> |                                                                    |                                                    |                                        |                    |
| Metoprolol tartrate  | 2.5-5 mg bolus over 2 min;<br>up to 3 doses                        | 25-200 mg, twice daily                             | 3-4 h                                  |                    |
| Metoprolol succinate | N/A                                                                | 50-400 mg daily or twice<br>daily in divided doses | 3-7 h                                  |                    |
| Atenolol             | N/A                                                                | 25-100 mg daily                                    | 6-7 h                                  | Renally eliminated |
| Bisoprolol           | N/A                                                                | 2.5-10 mg daily                                    | 9-12 h                                 |                    |
| Carvedilol           | N/A                                                                | 3.125-25 mg, twice daily                           | 7-10 h                                 |                    |
| Esmolol              | 500 µg/kg bolus over 1 min;<br>then 50-300 µg/kg/min               | N/A                                                | 9 min                                  |                    |
| Nadolol              | N/A                                                                | 10-240 mg daily                                    | 20-24 h                                |                    |
| Propranolol          | 1 mg over 1 min; repeat as<br>needed every 2 min; up to<br>3 doses | 10-40 mg, 3-4 times daily                          | IV: 2.4 h<br>Oral: 3-6 h<br>ER: 8-20 h |                    |

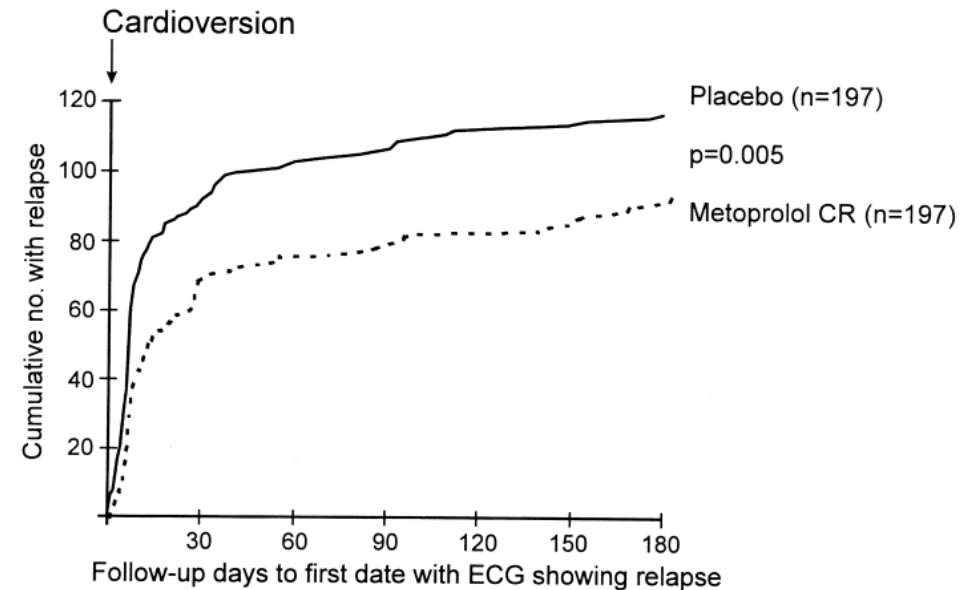
## Electrophysiology

# Use of Metoprolol CR/XL to Maintain Sinus Rhythm After Conversion from Persistent Atrial Fibrillation

A Randomized, Double-Blind, Placebo-Controlled Study

Volker Kühlkamp,\* Alexander Schirdewan,† Karl Stangl,‡ Michael Homberg,§ Matthias Ploch,|| Otto A. Beck¶

- P: AF post cardioversion
- I: Metoprolol
- C: Placebo
- O: Maintenance of sinus rhythm after cardioversion



**Figure 2.** Cumulative number of patients with relapse (atrial fibrillation/atrial flutter), subdivided by treatment group. The log-rank test of equality over treatment groups yielded a p value = 0.005; hence, the two treatment groups differ significantly with regard to the rate of relapse. The p value refers to the complete follow-up time (intention-to-treat approach).



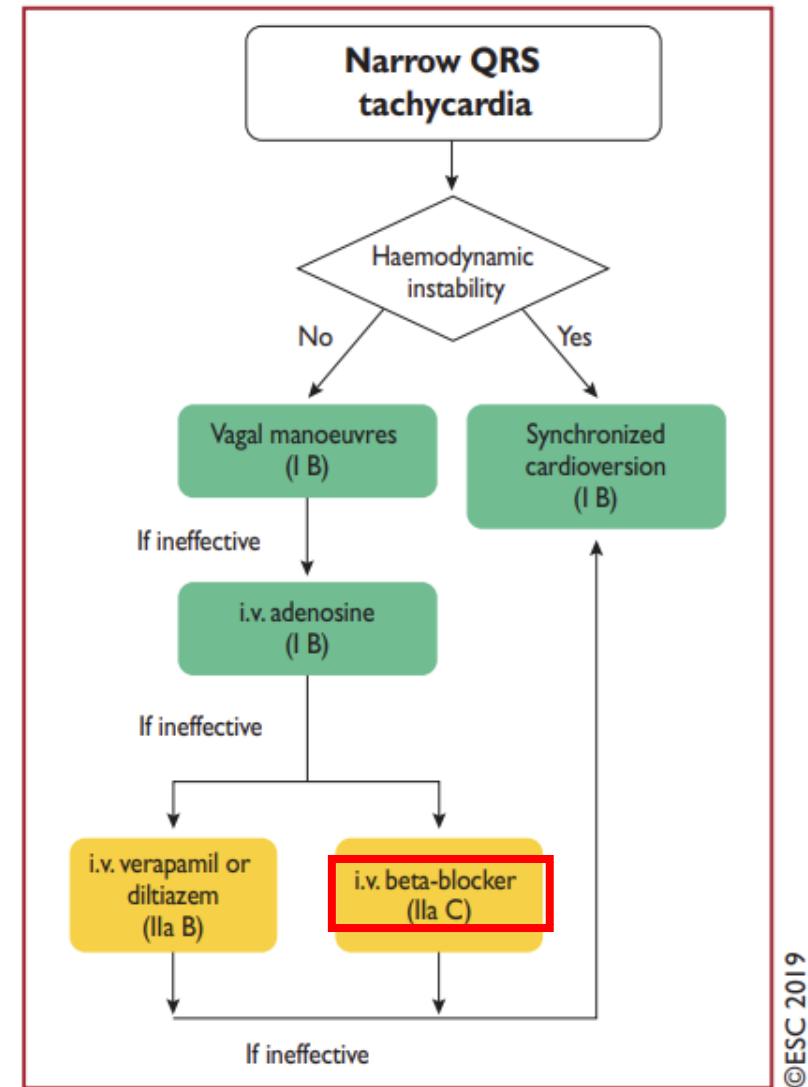
# *Supraventricular tachycardia*

# *Supraventricular tachycardia*

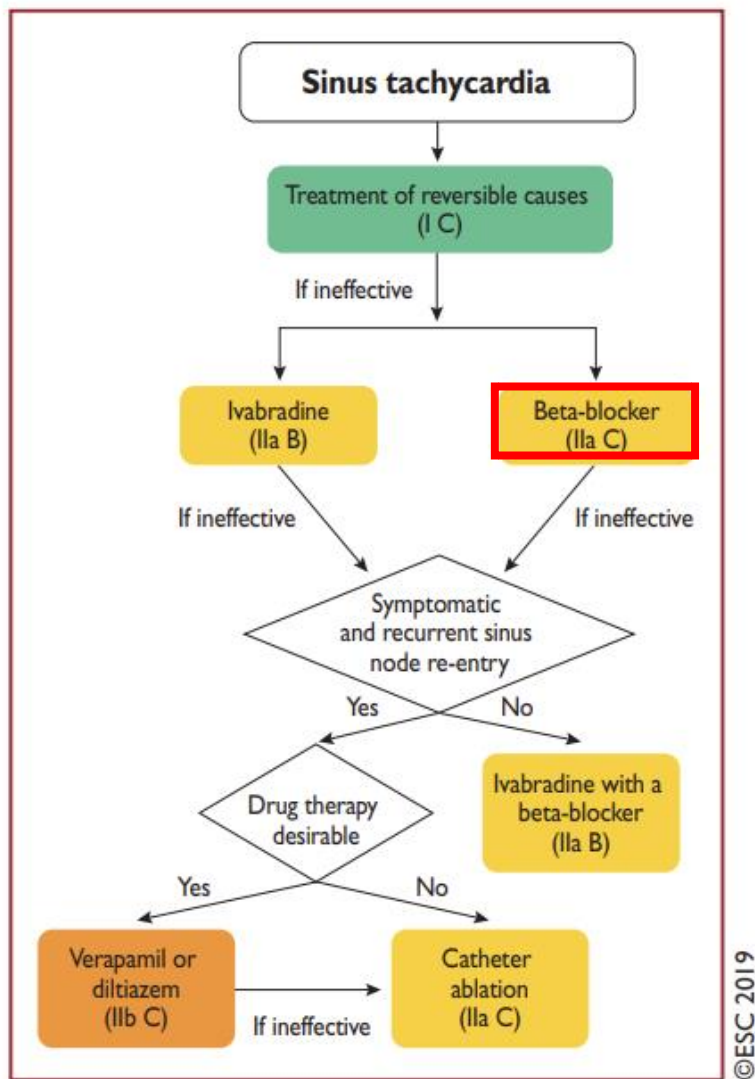
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- Mechanism
  - Decrease sympathetic tone
  - AV Nodal Blockade
- Contraindicated in Pre-excitation Syndromes
- Atenolol, Metoprolol are generally recommended
  - Cardio-selective BBs with extensively studied
- SVT treatment
  - Acute treatment
    - IV Beta blocker: Esmolol
  - Chronic therapy
    - Oral beta blocker: Metoprolol, Atenolol

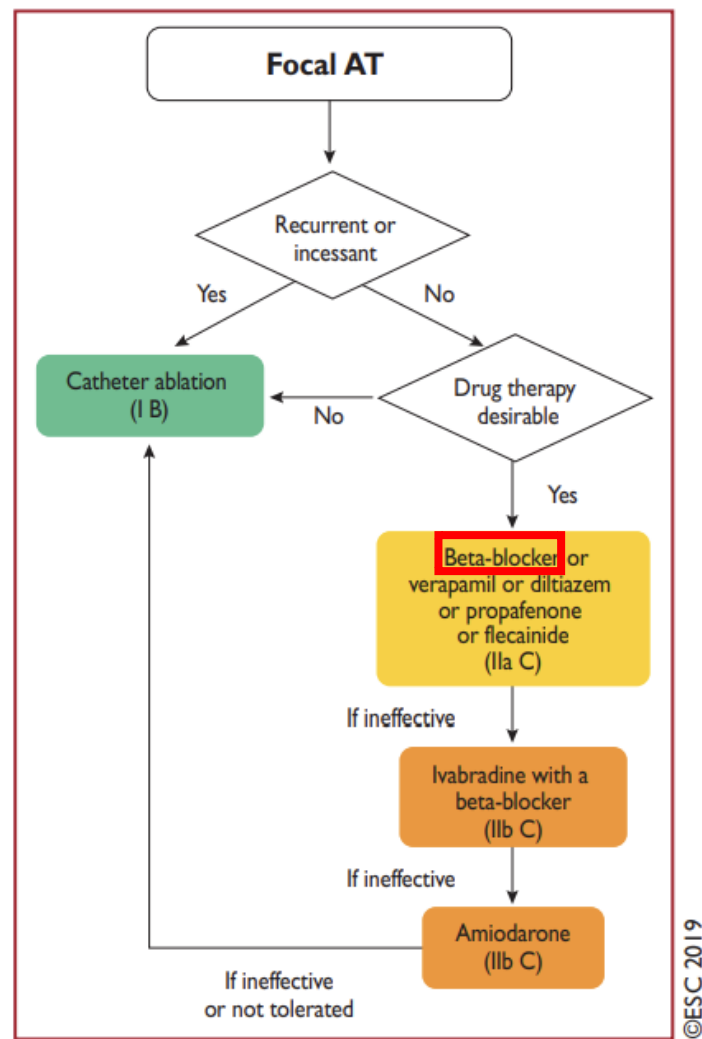
# Acute therapy of SVT



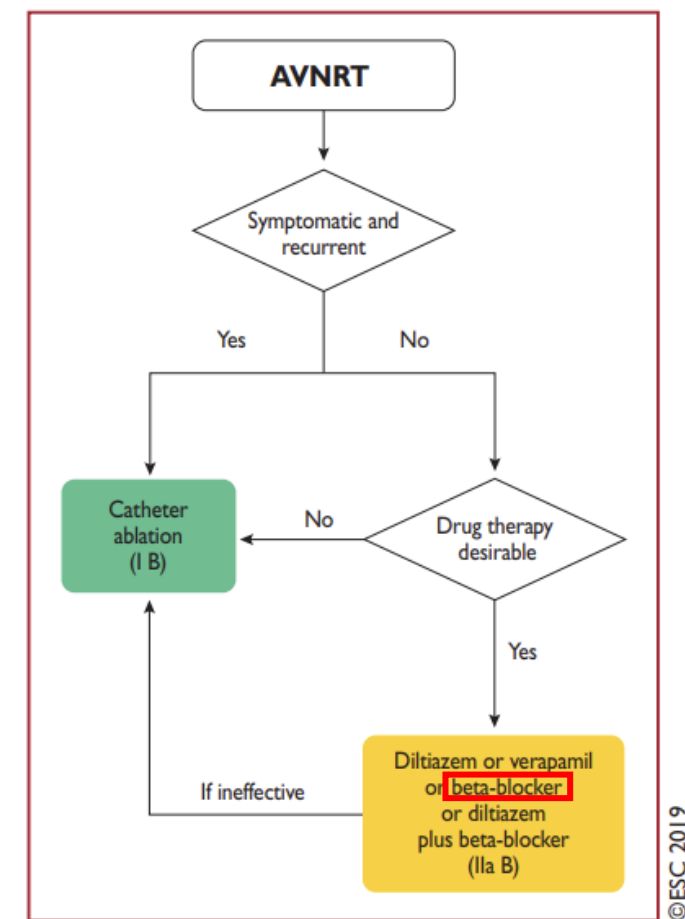
**Figure 4** Acute therapy of narrow QRS tachycardia in the absence of an established diagnosis.  
i.v. = intravenous.



**Figure 6** Therapy of sinus tachycardias.

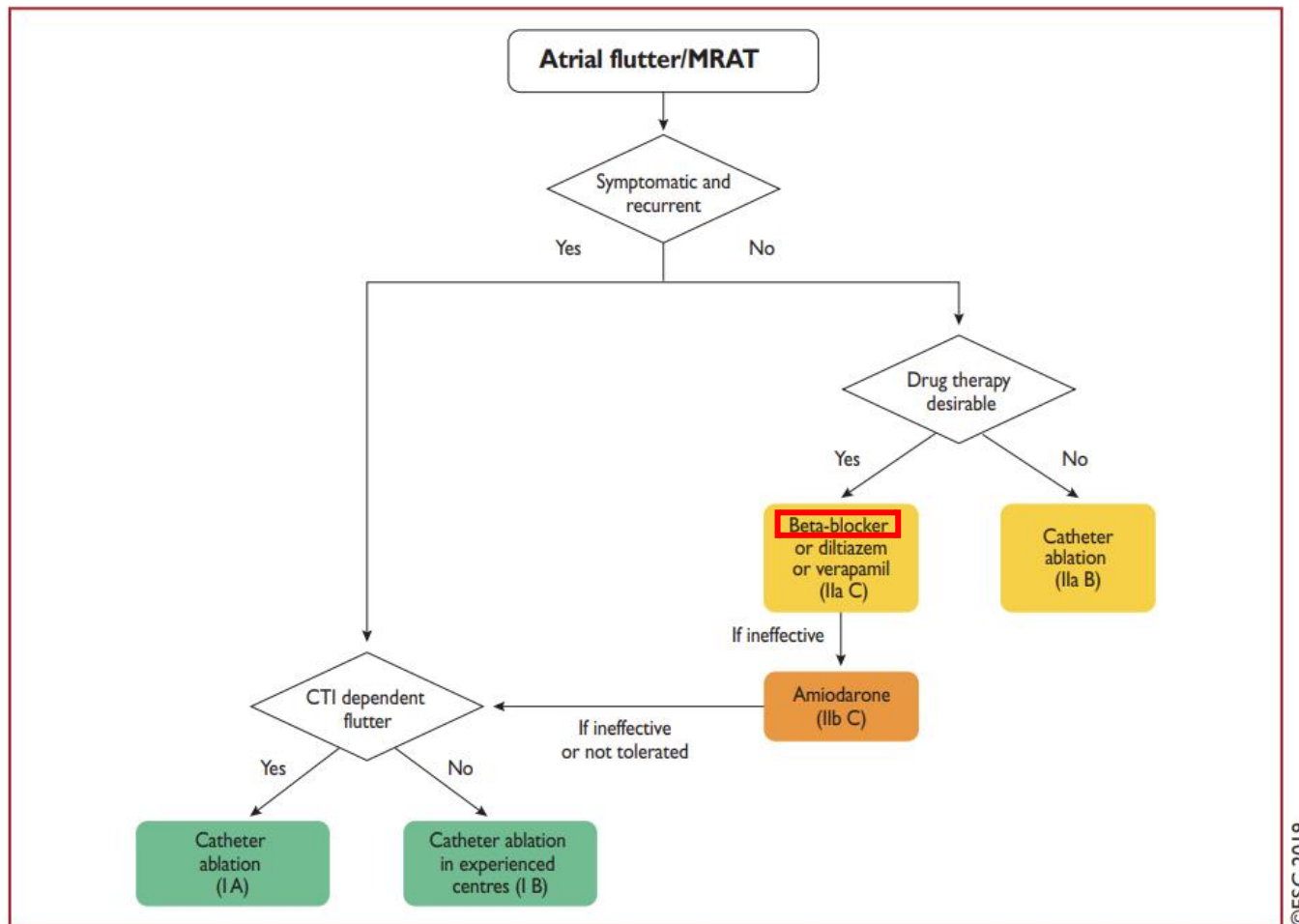


**Figure 9** Chronic therapy of focal atrial tachycardia.  
AT = atrial tachycardia.

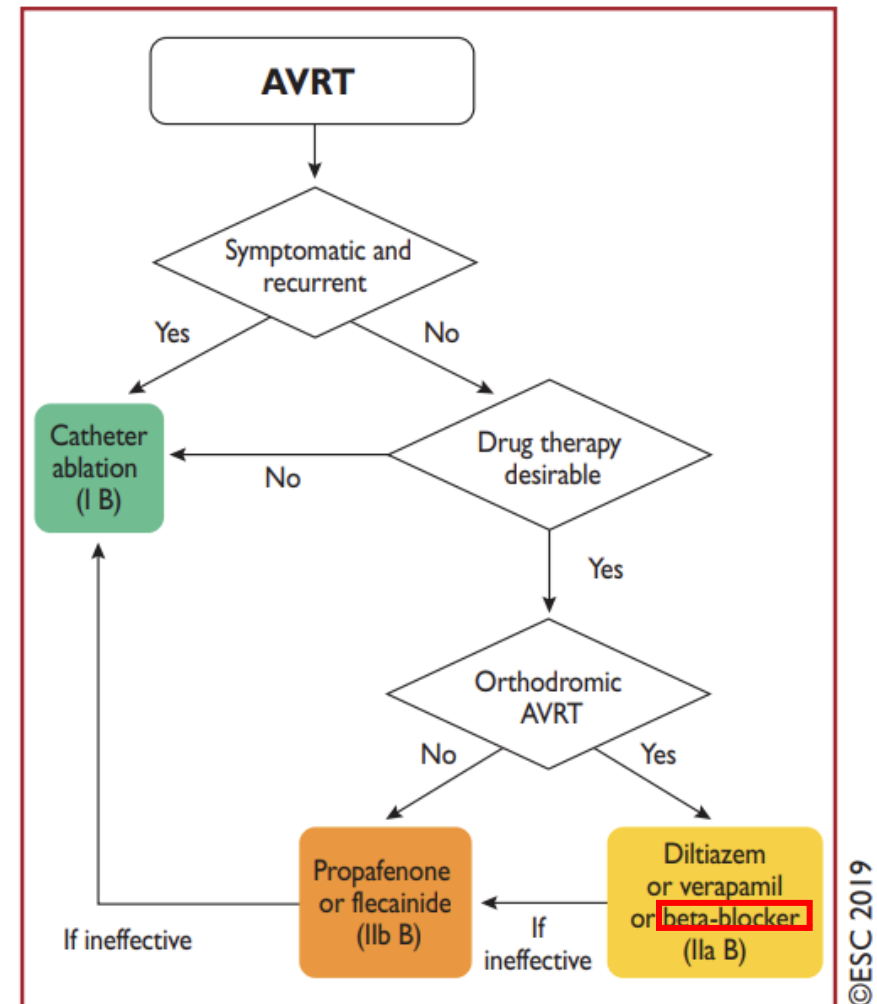


**Figure 15** Chronic therapy of atrioventricular nodal re-entrant tachycardia.

AVNRT = atrioventricular nodal re-entrant tachycardia.



**Figure 12** Chronic therapy of atrial flutter/macro-re-entrant atrial tachycardia. CTI = cavotricuspid isthmus; MRAT = macro-re-entrant atrial tachycardia.



**Figure 21** Chronic therapy of atrioventricular re-entrant tachycardia. AVRT = atrioventricular re-entrant tachycardia.

# Acute therapy for SVT

| Beta blockers       |                                |                                                                                 |                                                      |                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------|--------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Esmolol             | 500-mcg/kg IV bolus over 1 min | Infusion at 50–300 mcg/kg/min, with repeat boluses between each dosing increase | Hypotension, worsening HF, bronchospasm, bradycardia | <ul style="list-style-type: none"> <li>• AV block greater than first degree or SA node dysfunction (in absence of pacemaker)</li> <li>• Decompensated systolic HF</li> <li>• Hypotension</li> <li>• Cardiogenic shock</li> <li>• Reactive airway disease</li> <li>• Renal dysfunction</li> <li>• Drugs with SA and/or AV nodal–blocking properties</li> </ul>   |
| Metoprolol tartrate | 2.5–5.0-mg IV bolus over 2 min | Can repeat 2.5- to 5.0-mg IV bolus in 10 min, up to 3 doses                     | Hypotension, worsening HF, bronchospasm, bradycardia | <ul style="list-style-type: none"> <li>• AV block greater than first degree or SA node dysfunction (in absence of pacemaker)</li> <li>• Decompensated systolic HF</li> <li>• Hypotension</li> <li>• Reactive airway disease</li> <li>• Drugs with SA and/or AV nodal–blocking properties</li> </ul>                                                             |
| Propranolol         | 1 mg IV over 1 min             | Can repeat 1 mg IV at 2-min intervals, up to 3 doses                            | Hypotension, worsening HF, bronchospasm, bradycardia | <ul style="list-style-type: none"> <li>• AV block greater than first degree or SA node dysfunction (in absence of pacemaker)</li> <li>• Cardiogenic shock</li> <li>• Reactive airway disease</li> <li>• Decompensated HF</li> <li>• Hypotension</li> <li>• Hepatic or renal dysfunction</li> <li>• Drugs with SA and/or AV nodal–blocking properties</li> </ul> |

# Ongoing drug therapy for SVT

**Table 7. Ongoing Drug Therapy for SVT, Oral Administration\***

| Drug                               | Initial Daily Dose(s)                                            | Maximum Total Daily Maintenance Dose                                 | Potential Adverse Effects                            | Precautions (Exclude or Use With Caution) and Interactions                                                                                                                                                                                                                                                                                           |
|------------------------------------|------------------------------------------------------------------|----------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Beta blockers                      |                                                                  |                                                                      |                                                      |                                                                                                                                                                                                                                                                                                                                                      |
| Atenolol                           | 25–50 mg QD                                                      | 100 mg QD (reduced dosing in patients with severe renal dysfunction) | Hypotension, bronchospasm, bradycardia               | <ul style="list-style-type: none"> <li>• AV block greater than first degree or SA node dysfunction (in absence of pacemaker)</li> <li>• Decompensated systolic HF</li> <li>• Hypotension</li> <li>• Reactive airway disease</li> <li>• Severe renal dysfunction</li> <li>• Drugs with SA and/or AV nodal–blocking properties</li> </ul>              |
| Metoprolol tartrate                | 25 mg BID                                                        | 200 mg BID                                                           | Hypotension, bronchospasm, bradycardia               | <ul style="list-style-type: none"> <li>• AV block greater than first degree or SA node dysfunction (in absence of pacemaker)</li> <li>• Decompensated systolic HF</li> <li>• Hypotension</li> <li>• Reactive airway disease</li> <li>• Drugs with SA and/or AV nodal–blocking properties</li> </ul>                                                  |
| Metoprolol succinate (long-acting) | 50 mg QD                                                         | 400 mg QD                                                            | Hypotension, bronchospasm, bradycardia               | <ul style="list-style-type: none"> <li>• AV block greater than first degree or SA node dysfunction (in absence of pacemaker)</li> <li>• Decompensated systolic HF</li> <li>• Hypotension</li> <li>• Reactive airway disease</li> <li>• Drugs with SA and/or AV nodal–blocking properties</li> </ul>                                                  |
| Nadolol                            | 40 mg QD                                                         | 320 mg QD (reduced dosage with renal impairment)                     | Hypotension, bronchospasm, bradycardia               | <ul style="list-style-type: none"> <li>• AV block greater than first degree or SA node dysfunction (in absence of pacemaker)</li> <li>• Reactive airway disease</li> <li>• Cardiogenic shock</li> <li>• Decompensated HF</li> <li>• Hypotension</li> <li>• Renal dysfunction</li> <li>• Drugs with SA and/or AV nodal–blocking properties</li> </ul> |
| Propranolol                        | 30–60 mg in divided or single dose with long-acting formulations | 40–160 mg in divided or single dose with long-acting formulations    | Hypotension, worsening HF, bronchospasm, bradycardia | <ul style="list-style-type: none"> <li>• AV block greater than first degree or SA node dysfunction (in absence of pacemaker)</li> <li>• Reactive airway disease</li> <li>• Decompensated systolic HF</li> <li>• Hypotension</li> <li>• Drugs with SA and/or AV nodal–blocking properties</li> </ul>                                                  |



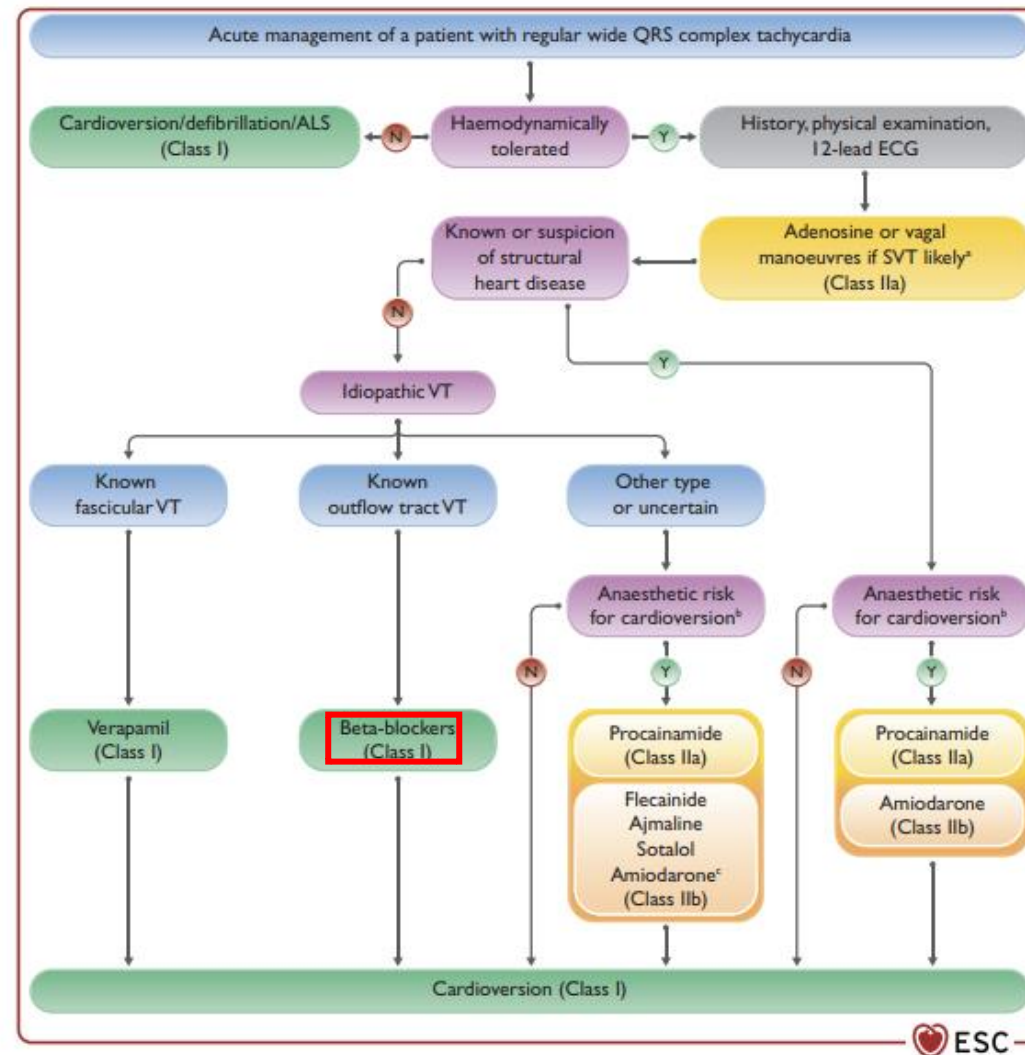
# *Ventricular arrhythmias*

# *Ventricular arrhythmias*

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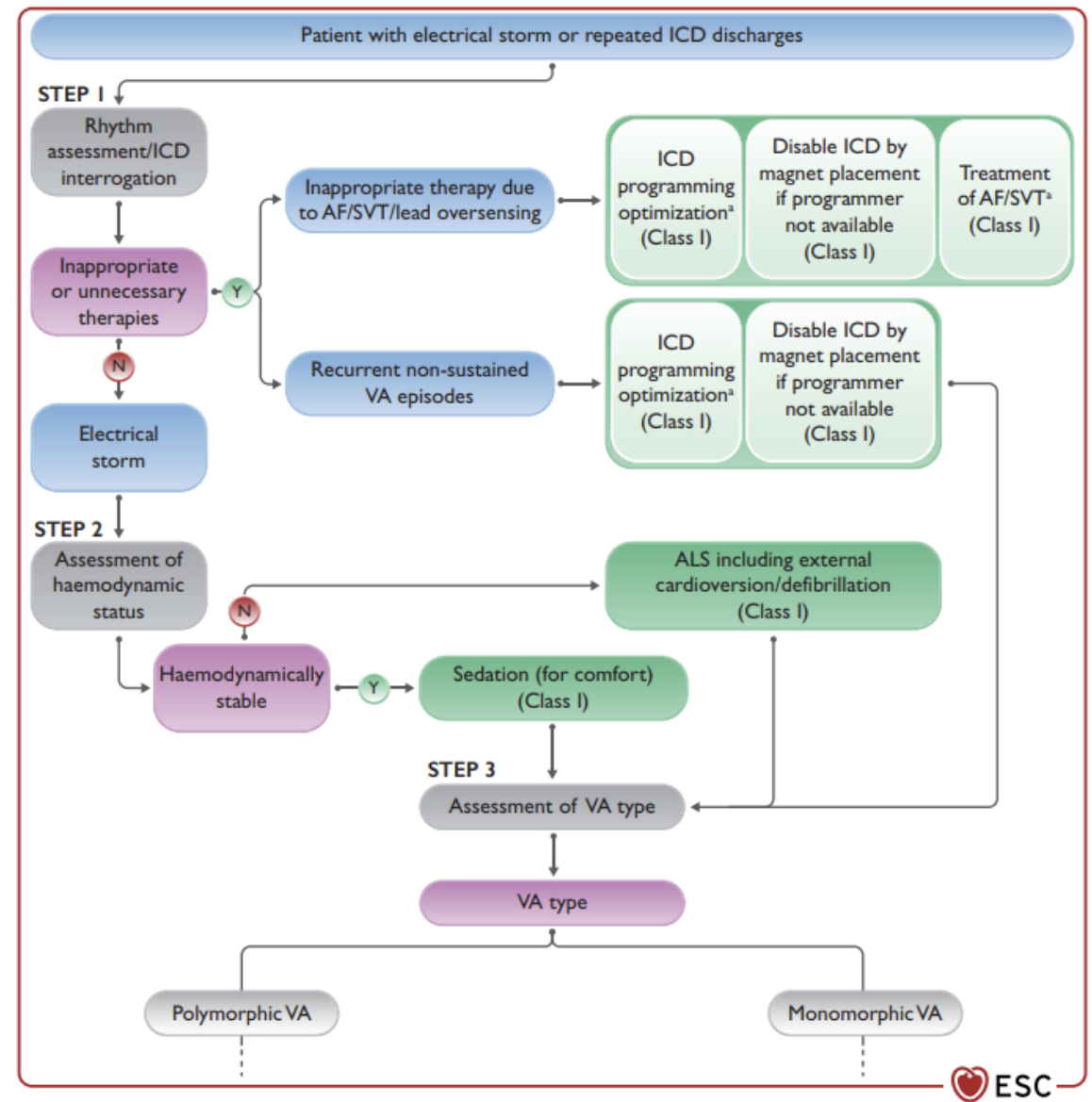
- Beta-blocker inhibit sympathetic activity
  - Very well respond in sympathetic activated VAs
- Treatment and prevention of VT/Symptomatic PVC
- 1<sup>st</sup> line Medical treatment for VT from specified cause
  - Long QT syndrome
  - Outflow tract VT
  - Catecholaminergic polymorphic ventricular tachycardia
- Most clinical trial use metoprolol, bisoprolol, carvedilol, or propranolol

# ESC Ventricular arrhythmias guideline 2022

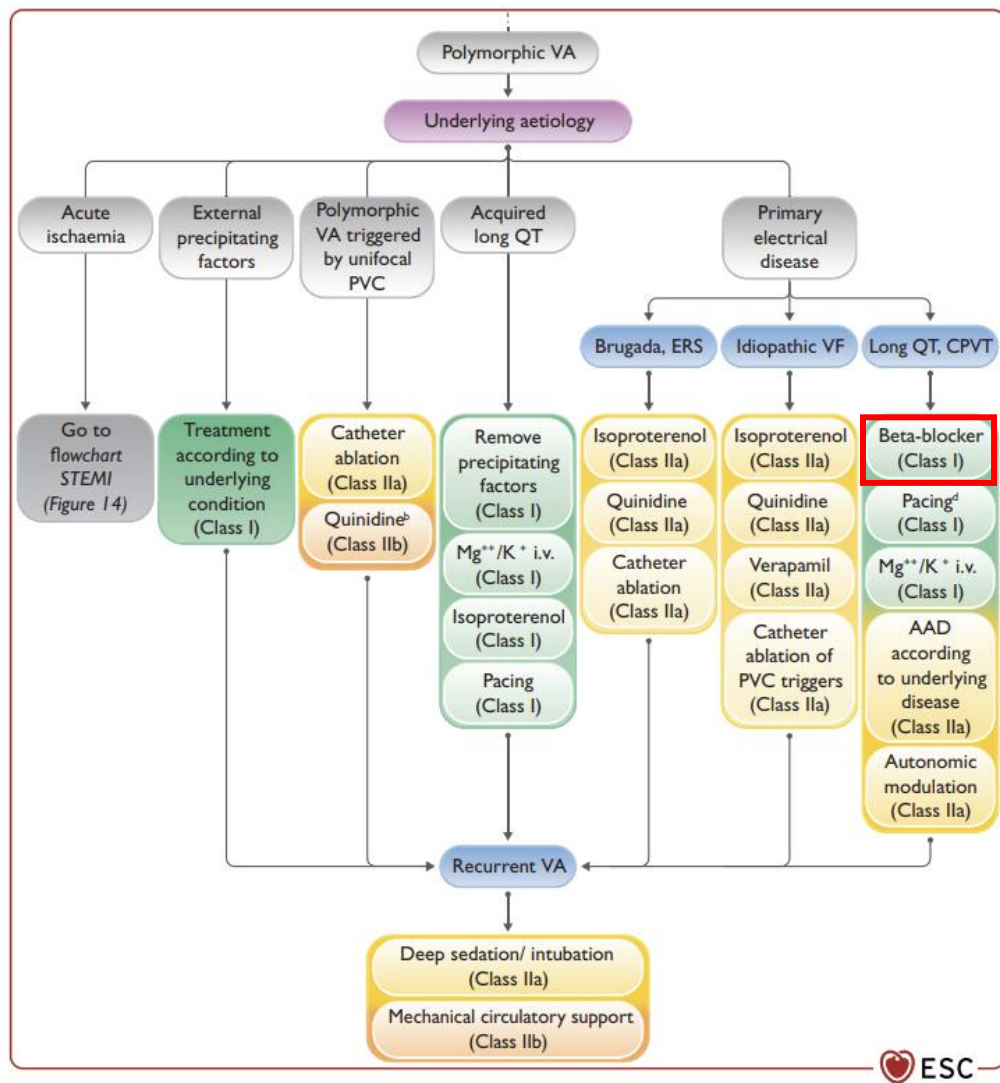


**Figure 9** Algorithm for the acute management of regular wide QRS complex tachycardia. ALS, advanced life support; ECG, electrocardiogram; N, No; SVT, supraventricular tachycardia; VT, ventricular tachycardia; Y, Yes. <sup>a</sup>Besides SVT, adenosine may also terminate idiopathic VT, which then indicates triggered activity as the mechanism underlying the arrhythmia. <sup>b</sup>Benefit of cardioversion should be weighed against risks related to anaesthesia/sedation. <sup>c</sup>Considering limited availability of the other anti-arrhythmic drugs.

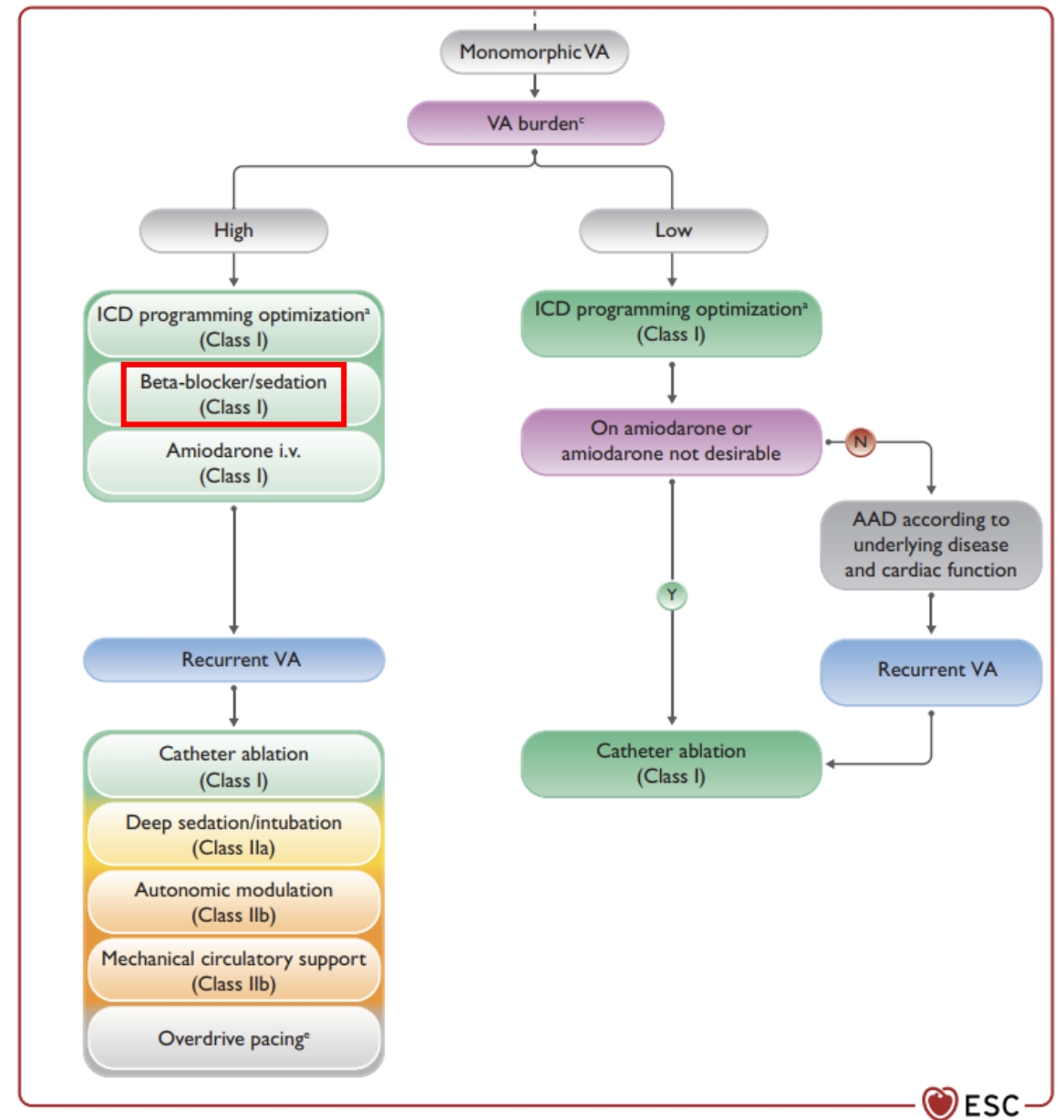
# Electrical storm treatment

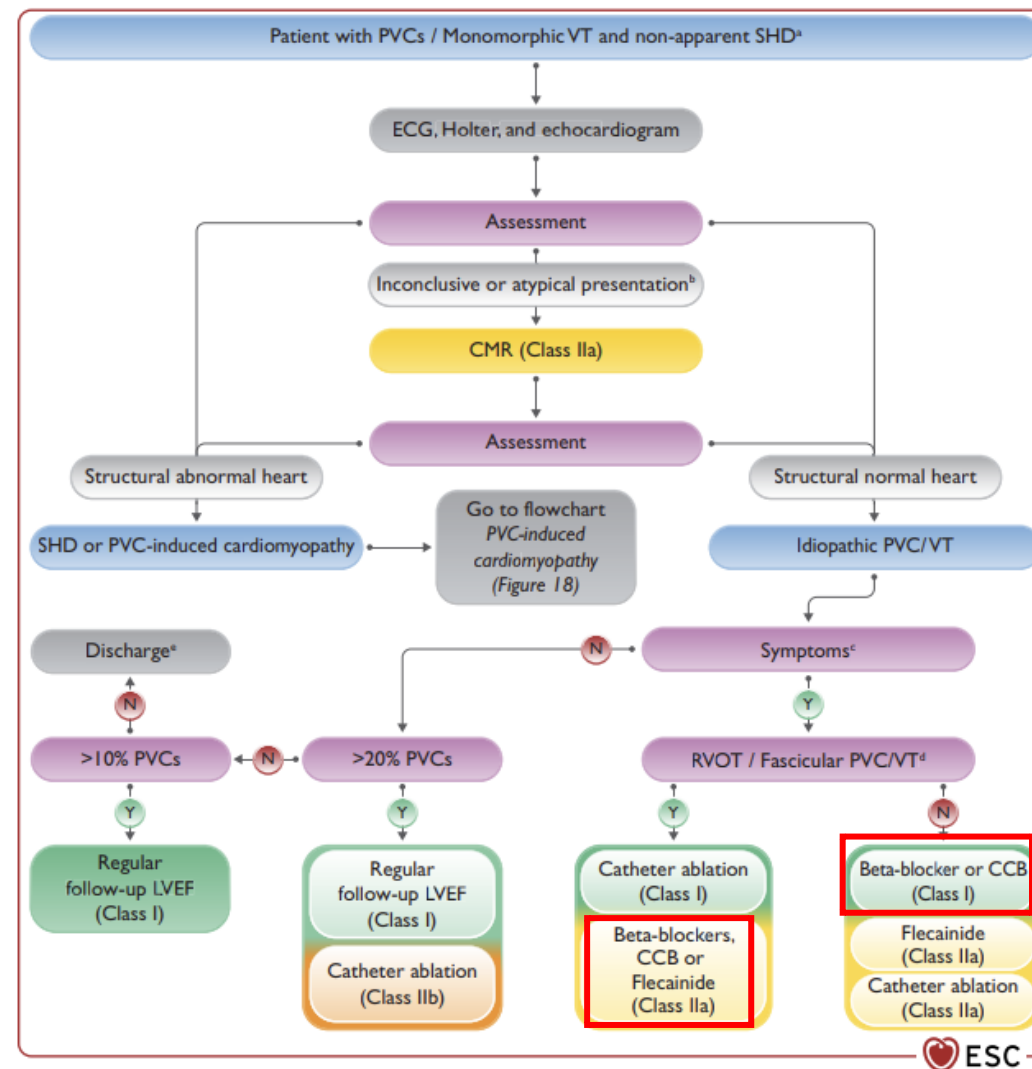


**Figure 11 Part One.** Management of patients with electrical storm or repeated implantable cardioverter defibrillator discharges.



**Figure 11 Part Two.** Management of patients with electrical storm or repeated implantable cardioverter defibrillator discharges.





**Figure 17** Algorithm for the management of patients with idiopathic premature ventricular complexes/ventricular tachycardia and non-apparent structural heart disease. CCB, calcium channel blocker; CMR, cardiac magnetic resonance; ECG, electrocardiogram; LVEF, left ventricular ejection fraction; N, No; PVC, premature ventricular complex; RVOT, right ventricular outflow tract; SHD, structural heart disease; VT, ventricular tachycardia; Y, Yes. <sup>a</sup>Non-apparent SHD is defined by lack of significant abnormalities in physical examination, basal ECG, and echocardiogram. <sup>b</sup>Atypical presentation: e.g. older age, right bundle branch block morphology, sustained monomorphic VT consistent with re-entry. <sup>c</sup>Symptoms should be relevant and related to PVC/VT. <sup>d</sup>Origin suspected by ECG or confirmed during electrophysiological evaluation. <sup>e</sup>Consider re-evaluation in case of new symptoms or changes in patient clinical condition.

# AHA ventricular arrhythmias guideline 2017

|                                                                                                                                                  |                        |                                     |                                                           |                                                                                             |                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Atenolol (II)<br>PO: 25–100 mg qd<br>or bid                                                                                                      | VT, PVC, ARVC,<br>LQTS | Beta 1                              | Sinus rate slowed<br>AV nodal refractoriness<br>increased | $t_{1/2}$ : 6–7 h<br>(prolonged with renal<br>impairment)<br>Metab: H<br>Excr: F 50%, U 40% | Cardiac: Bradycardia,<br>hypotension, HF, AVB<br>Other: Dizziness, fatigue,<br>depression, impotence                                       |
| Bisoprolol (II)<br>PO: 2.5–10 mg once<br>daily                                                                                                   | VT, PVC                | Beta 1 receptor                     | Sinus rate slowed<br>AV nodal refractoriness<br>increased | $t_{1/2}$ : 9–12 h<br>Metab: H<br>Excr: U                                                   | Cardiac: Chest pain,<br>bradycardia, AVB<br>Other: Fatigue, insomnia,<br>diarrhea                                                          |
| Carvedilol (II)<br>PO: 3.125–25 mg q<br>12 h                                                                                                     | VT, PVC                | Beta 1 and 2 receptors,<br>Alpha    | Sinus rate slowed<br>AV nodal refractoriness<br>increased | $t_{1/2}$ : 7–10 h<br>Metab: H<br>Excr: F                                                   | Cardiac: Bradycardia,<br>hypotension, AVB, edema,<br>syncope<br>Other: Hyperglycemia,<br>dizziness, fatigue, diarrhea                      |
| Esmolol (II)<br>IV: 0.5 mg/kg bolus,<br>0.05 mg/kg/min                                                                                           | VT                     | Beta 1 receptor                     | Sinus rate slowed<br>AV nodal refractoriness<br>increased | $t_{1/2}$ : 9 min<br>Metab: RBC esterases<br>Excr: U                                        | Cardiac: Bradycardia,<br>hypotension, HF, AVB<br>Other: Dizziness, nausea                                                                  |
| Metoprolol (II)<br>IV: 5 mg q 5 min up to<br>3 doses<br>PO: 25–100 mg<br>Extended release qd<br>or q 12 h                                        | VT, PVC                | Beta 1 receptor                     | Sinus rate slowed<br>AV nodal refractoriness<br>increased | $t_{1/2}$ : 3–4 h<br>Metab: H<br>Excr: U                                                    | Cardiac: Bradycardia,<br>hypotension, AVB<br>Other: Dizziness, fatigue,<br>diarrhea, depression, dyspnea                                   |
| Propranolol (II)<br>IV: 1–3 mg q 5 min to a<br>total of 5 mg<br>PO: Immediate<br>release 10–40 mg q<br>6 h; Extended release<br>60–160 mg q 12 h | VT, PVC, LQTS          | Beta 1 and 2 receptors,<br>$I_{Na}$ | Sinus rate slowed<br>AV nodal refractoriness<br>increased | $t_{1/2}$ : Immediate release<br>3–6 h<br>Extended release 8–10 h<br>Metab: H<br>Excr: U    | Cardiac: Bradycardia,<br>hypotension, HF, AVB<br>Other: Sleep disorder, dizziness,<br>nightmares, hyperglycemia,<br>diarrhea, bronchospasm |



*Migraine*

# *Migraine*

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- Betablockers can reduce neuronal excitability
- Recommended medication for secondary prevention of migraine
- Recommended betablocker > BB with high lipophilicity
  - Propranolol, Metoprolol (Level A)
- Atenolol has clinical benefit evidence (Level B)
  - Low lipophilicity

## ตารางที่ 10 กลุ่มยาปิดกั้นตัวรับเบต้า

| ยา             | การเริ่มให้ยา/<br>การเพิ่มระดับยา                                                      | ขนาดยา<br>ที่แนะนำ                                                                      | ภาวะที่ควรหลีกเลี่ยง<br>หรือห้ามใช้                                                                     | โรคร่วม<br>ที่เหมาะสม          | ผลข้างเคียง                                                                                                                                             |
|----------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Beta blockers: |                                                                                        |                                                                                         |                                                                                                         |                                |                                                                                                                                                         |
| Propranolol    | 20-40 มก.<br>วันละ 2 ครั้ง;<br>เพิ่มครั้งละ 20 มก.<br>วันละ 2 ครั้ง<br>ทุก 1-2 สัปดาห์ | 80-160 มก./วัน<br>(แบ่งเป็น 2 เวลา<br>หรือวันละครั้ง<br>กรณียาออกฤทธิ์<br>ยาว)          | หอบหืด, heart block,<br>ภาวะหัวใจล้มเหลว,<br>ความดันโลหิตต่ำ,<br>หัวใจเต้นช้า, โรคหลอดเลือด<br>ส่วนปลาย | ความดัน<br>โลหิตสูง,<br>angina | อ่อนเพลีย, exercise<br>intolerance, หัวใจ<br>เต้นช้า, หัวใจล้มเหลว,<br>ความดันโลหิตต่ำ,<br>หลอดเลือดหัวใจ,<br>เสื่อมสมรรถภาพ<br>ทางเพศ, รบกวน<br>การนอน |
| Metoprolol     | 50 มก. วันละ 2 ครั้ง<br>(วันละ 1 ครั้ง กรณีเป็น<br>sustain release<br>form)            | 100-200 มก./วัน<br>(แบ่งวันละ 2 ครั้ง<br>หรือ 1 ครั้ง/วัน<br>กรณีเป็นยา<br>ออกฤทธิ์ยาว) | (Raynaud's disease),<br>type 1 DM, ซึมเศร้า,<br>sexual dysfunction                                      |                                |                                                                                                                                                         |
| Atenolol       | 25 มก. วันละ 1 ครั้ง                                                                   | 50-100 มก./วัน                                                                          |                                                                                                         |                                |                                                                                                                                                         |



# *Hyperthyroidism*

# *Hyperthyroidism*

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- Adrenergic Symptom Control
  - Control Tachycardia, Palpitation, Tremor
- Propranolol is generally recommended due to additional ability to inhibit peripheral T<sub>4</sub> to T<sub>3</sub> conversion
- Esmolol is recommended for acute heart rate control in thyroid storm
- Cardio-selective betablocker may use in asthma/COPD

TABLE 4. BETA-ADRENERGIC RECEPTOR BLOCKADE IN THE TREATMENT OF THYROTOXICOSIS

| <i>Drug<sup>a</sup></i> | <i>Dosage</i>                 | <i>Frequency</i>  | <i>Considerations</i>                                                                                                                                                                               |
|-------------------------|-------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Propanolol <sup>b</sup> | 10–40 mg                      | 3–4 times per day | Nonselective $\beta$ -adrenergic receptor blockade<br>Longest experience<br>May block T <sub>4</sub> to T <sub>3</sub> conversion at high doses<br>Preferred agent for nursing and pregnant mothers |
| Atenolol                | 25–100 mg                     | 1–2 times per day | Relative $\beta$ -1 selectivity<br>Increased compliance<br>Avoid during pregnancy                                                                                                                   |
| Metoprolol <sup>b</sup> | 25–50 mg                      | 2–3 times per day | Relative $\beta$ -1 selectivity                                                                                                                                                                     |
| Nadolol                 | 40–160 mg                     | 1 time per day    | Nonselective $\beta$ -adrenergic receptor blockade<br>Once daily<br>Least experience to date<br>May block T <sub>4</sub> to T <sub>3</sub> conversion at high doses                                 |
| Esmolol                 | IV pump 50–100 $\mu$ g/kg/min |                   | In intensive care unit setting of severe thyrotoxicosis or storm                                                                                                                                    |

<sup>a</sup>Each of these drugs has been approved for treatment of cardiovascular diseases, but to date none has been approved for the treatment of thyrotoxicosis.

<sup>b</sup>Also available in once daily preparations.  
T<sub>4</sub>, thyroxine.



# *Portal hypertension*

# *Portal hypertension*

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- Beta-blocker is recommended in clinically significant portal hypertension (CSPH) in cirrhosis
  - CSPH: Hepatic venous pressure gradient  $\geq 10$  mm Hg
    - History of variceal bleeding, High risk varices found, Platelet  $< 100,000$ - $150,000$  / $\mu$ L, Liver stiffness  $\geq 20$ – $25$  kPa
- Mechanism of portal pressure reduction
  - Decrease cardiac output (Beta1)
  - Decrease splanchnic arterial vasoconstriction (Beta2)
  - Intrahepatic vasodilation (Alpha1 block)
- Recommend BB: Carvedilol, Propranolol
- In refractory ascites betablocker associated hypotension risk may  $>$  benefit
  - Especially carvedilol

**TABLE 3** Nonselective beta-blockers used in portal hypertension

| Therapy     | Mechanism of action                                                                                                   | Starting dose        | Titration                                          | Maximal dose                                                              | Goal                                                                   | Common adverse effects                                                | Maintenance                                                                               |
|-------------|-----------------------------------------------------------------------------------------------------------------------|----------------------|----------------------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Propranolol | Decreased cardiac output; caused by decreased heart rate and contractility from beta-1 adrenergic blockade, plus      | 20–40 mg twice daily | Increase the dose every 2–3 d until treatment goal | Without ascites: 320 mg/day; with ascites: 160 mg/day                     | HR of 55–60 bpm if tolerated; SBP should be maintained $\geq 90$ mm Hg | Fatigue, bradycardia, dyspnea, orthostasis, hypotension, constipation | Indefinitely or until TIPS or liver transplant. No indication for routine upper endoscopy |
| Nadolol     | Splanchnic arterial vasoconstriction; caused by beta-2 blockade leading to unopposed alfa-adrenergic vasoconstriction | 20–40 mg at bedtime  |                                                    | Without ascites: 160 mg/day; with ascites: 80 mg/day                      |                                                                        |                                                                       |                                                                                           |
| Carvedilol  | Above plus decreased intrahepatic vascular resistance; caused by anti-alpha-adrenergic activity                       | 6.25 mg once daily   | Increase to 6.25 mg twice daily after 3 d          | 12.5 mg/day (higher doses could be considered for nonhepatic indications) | No HR goal; SBP should be maintained $\geq 90$ mm Hg                   |                                                                       |                                                                                           |

**Abbreviations:** SBP, spontaneous bacterial peritonitis; HR, heart rate.



# *Essential tremor*

# *Essential tremor*

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- Beta blockers Inhibit beta-adrenergic receptors in peripheral muscles
  - May modulating neuronal firing in brain
- Propranolol is most widely used
  - Strongest evidence based
  - Most frequently and successfully to treat ET

# *Beta blocker indication*

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- Essential hypertension: Carvedilol, Nebivolol
- Hypertensive emergency: Labetalol
- HFrEF: Carvedilol, Nebivolol, Metoprolol, Bisoprolol
- Coronary artery disease: ?
- Acute treatment in arrhythmia: Esmolol
- AF: Metoprolol, Atenolol, Bisoprolol, Carvedilol
- SVT: Metoprolol, Atenolol, Bisoprolol
- VT/PVC: Metoprolol, Bisoprolol, Carvedilol, Propranolol
- Migraine: Propranolol, Metoprolol
- Hyperthyroidism: Propranolol
- Essential tremor: Propranolol
- Portal hypertension: Carvedilol, Propranolol

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